KPS PUBLIC SCHOOLS • STUDENT TECH BENCH PROGRAM

CTL Chromebook NL73

Solder-Free Repair Guide



Covers NL73 and NL73-8 clamshells. NL73CT differences are flagged.

Not for NL73T / NL73TW / NL73TWB convertibles.

In this guide

1 Read first: safety, tools, basic skills 2

2 Know the machine: specs, anatomy, parts list 9

3 Triage: symptom → repair finder 12

4 Software repairs (S1–S3) 13

5 Hardware repairs (R1–R19) 17

6 Out of scope: send to CTL / lead tech 47

7 Sources 48

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Procedures follow CTL's NL72 Self-Repair Manual, which CTL designates as the reference for the NL73, plus CTL's NL73 specs and parts catalog. Confirm NL73-specific details on the first teardown.

1 Read first

This guide covers every NL73 repair the Tech Bench can do with hand tools: swapping a part, reseating a cable, or fixing software. Anything that needs a soldering iron, hot air, or board-level work is listed in Section 6 and goes to CTL or the district lead tech instead.

How this guide was built. CTL's support site lists the NL72 Self-Repair Manual as the reference for the NL73. The screw counts and step order here come from that manual. Parts come from CTL's NL73 catalog, and specs from CTL's NL73 spec sheet. Because the manual was written for the NL72, some repairs have a notes box. Confirm the counts the first time you do each repair on an NL73.

Which NL73 is on your bench? Read the model on the D-cover label. This guide is for the **NL73** (4 GB) and **NL73-8** (8 GB) clamshells. The **NL73CT** (touch clamshell) shares most parts except the screen. **NL73T, NL73TW, and NL73TWB** are convertibles with different keyboards, covers, panels, and hinges. CTL warns that some NL72T parts don't fit the NL73T. Don't use this guide for them.

Bench levels

Level	Who may do it	Typical repairs
LEVEL 1	Any trained student	Software resets, single keys, camera caps, base cover, battery
LEVEL 2	Student with supervisor sign-off on this repair	Display, bezel, cables, USB board, Wi-Fi card, touchpad, camera, hinges, speakers
LEVEL 3	Lead tech or adult supervisor only	Top assembly (keyboard) swap, A cover, liquid damage, mainboard swap

Before you touch any device

1. ☐ Check the asset tag and look the device up in the district inventory. Note any open ticket.
2. ☐ Ask the lead tech whether the unit is still under CTL warranty or an accidental-damage plan. CTL's manual says damage from careless or unsuccessful self-repair isn't covered by the standard warranty, so warranty units may be better sent to CTL.
3. ☐ Photograph all four sides of the device before you open it.
4. ☐ Record the symptom in the user's own words on the ticket.
5. ☐ Warn the user that a reset erases local files (Downloads). Cloud files in Drive are safe.

Safety rules

- **Power off, unplug, remove the microSD card and all cables** (CTL's pre-disassembly steps), then disconnect the battery before any work under the base cover.
- **Lithium batteries:** never puncture, bend, or pry against the battery. A swollen, hot, hissing, or smoking battery is a Level 3 stop. Move it to the metal battery bin or sand and get an adult.
- **ESD:** CTL's manual calls for a wrist grounding strap and conductive mat. Touch boards only by the edges.
- **Ribbon cables:** CTL warns the touchpad and USB-board FFCs can be damaged if pulled while their connectors are locked. Always unlock first.
- **Glass:** wear safety glasses when removing a cracked LCD.
- **Only use CTL-certified parts**, as CTL's manual requires.

Tools and supplies

Item	Use	Notes
Phillips #0 and #1 screwdrivers	All case and board screws	CTL's recommended set
Small flathead screwdriver	Releasing some latches	CTL's recommended set. Never on board components.
Flat plastic pry tool / spudger	Base cover, bezel, connectors	CTL's recommended set
Plastic tweezers	FFC ribbon cables, antenna leads	CTL's recommended set
Wrist grounding strap + conductive mat	Static protection	CTL's recommended set
Magnetic screw mat or labeled tray	Keeping screws in order	Base cover alone has 9 screws
90%+ isopropyl alcohol + soft brush	Cleaning contacts, spill residue	
Kapton tape	Re-taping cables and the keyboard-plate mylar	
Recovery USB or SD card, 8 GB+	ChromeOS recovery	Made with Chromebook Recovery Utility
Known-good USB-C charger (45 W)	Testing power and charging	CTL 45 W USB-C adapter
Safety glasses	Cracked-screen removal	
Fire-safe battery bin or sand bucket	Damaged batteries	Required at every bench

Basic skills (start here if you're new)

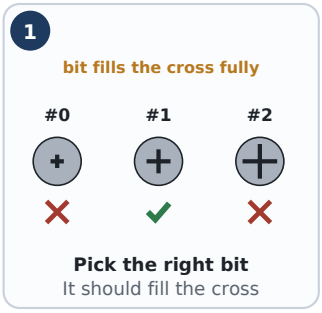
Every repair in this guide uses the same handful of skills. Learn these six first and the rest of the guide is just *which screws* and *which cable*. If a word in a repair is unfamiliar, look it up in the word list at the end of this chapter. Hard words in the steps are also explained in grey the first time they appear.

How a repair page works

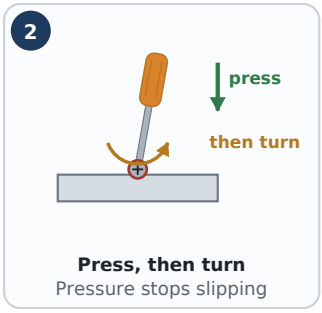
Part of the page	What it tells you
Coloured bar with the code (R5) and LEVEL	What the repair is and who may do it. Level 1: any trained student. Level 2: with a supervisor's sign-off. Level 3: lead tech only.
Fixes / Part / Time / Tools	Read this first. If you don't have the part or the tools, stop and get them before you open anything.
In plain words	One or two sentences on what you're about to do and why.
At a glance (numbered pictures)	The idea of the repair in four pictures. They show the idea, not your exact model.
Where it is on this model	A map of the inside with the part highlighted.
Step by step (☐ boxes)	The exact steps. Tick each box as you go so you never lose your place.
Common mistakes / If something goes wrong	The traps other students fall into, and what to do when a step doesn't work.
Verify before closing the ticket	The checks that prove the repair worked. Never skip them.
First-teardown notes	Write in what you found the first time (real screw counts, tape positions). It helps the next student.

Skill 1: Screws

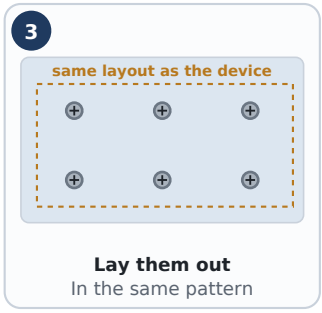
Most damage on a repair bench comes from screws: stripped heads, screws in the wrong hole, or a long screw driven through a board. Go slowly; screws are tiny and soft.



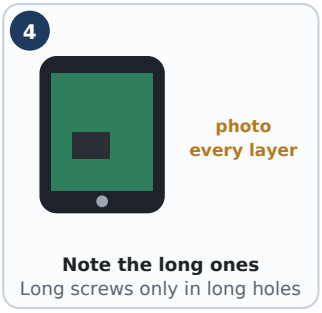
Pick the right bit
It should fill the cross



Press, then turn
Pressure stops slipping



Lay them out
In the same pattern

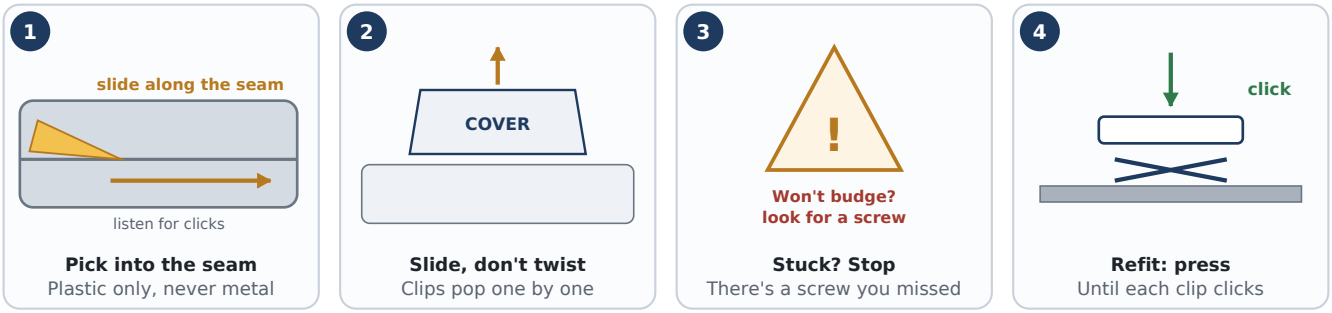


Note the long ones
Long screws only in long holes

- Most Chromebook screws take a **Phillips #0 or #1** bit. If the bit wobbles or slips, try the other size.
- Turn **left to loosen** (counter-clockwise), right to tighten. Tighten until it stops, then stop. "Snug" is enough: over-tightening cracks plastic.
- **Captive screws** stay in the cover. Loosen them until they spin freely; they won't come out.
- If a screw head starts to strip (the cross rounds off), stop and get the lead tech. Don't keep turning.

Skill 2: Opening clips

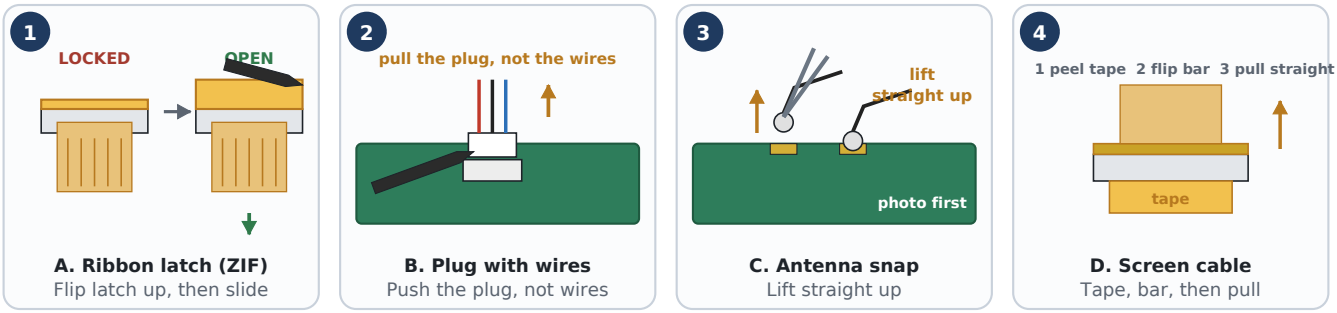
Covers and bezels are held by small plastic clips as well as screws. You release them by sliding a plastic pick along the seam, not by pulling.



- A click or pop is normal. A crack or creak means stop and look for a screw.
- Start where the guide says (usually a corner) and work around slowly.
- Never pry near the battery or against the screen glass.

Skill 3: The four kinds of connectors

Inside every Chromebook, parts plug into the main board in only four ways. Learn these and you can unplug anything safely.



Type	Looks like	How to unplug	How to plug back in
A. Ribbon latch (ZIF)	Flat, thin, see-through cable with gold stripes, in a slot with a small hinged flap	Flip the flap up with a fingernail or spudger. The ribbon slides out with no force.	Gold stripes facing the same way as before. Push it in straight and all the way, then close the flap.
B. Plug with wires	Small white plug with coloured wires (battery, speakers, fan)	Push the plug body out with a spudger, gently rocking. Never pull the wires.	Line it up, press straight in until it seats flat.
C. Antenna snap	Tiny round gold button on a thin black or white wire	Lift straight up with plastic tweezers or a fingernail.	Hold it right over the post and press straight down until it snaps.
D. Screen cable	Wider connector, often under tape, with a metal bar or pull tab	Peel the tape, flip up the metal bar (or pull the tab), then pull straight out.	Push straight in, flip the bar down, put the tape back.

Battery first, battery last. Unplug the battery (type B) before you unplug anything else, and plug it back in only after everything else is connected.

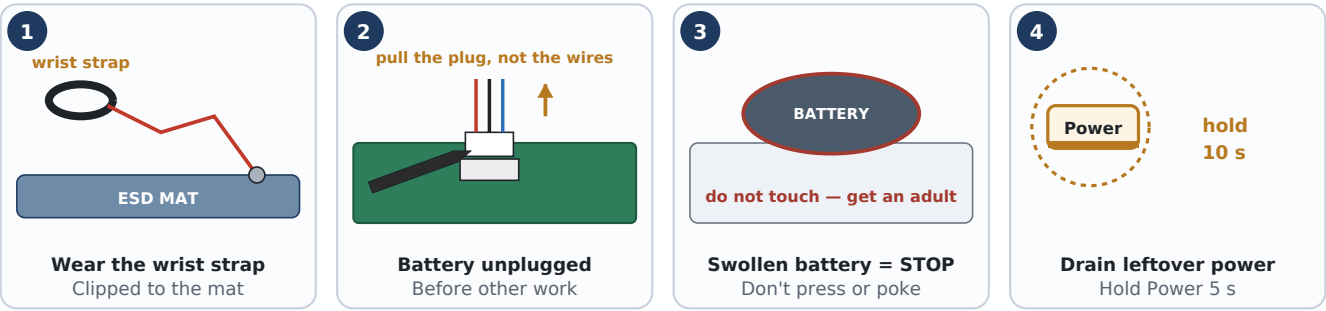
Skill 4: Tape, Mylar and adhesive

Many parts are held by tape as well as screws. Some tape is meant to be reused; some must be replaced.



- **Mylar** is thin plastic sheet that insulates parts. Put it back where it was: it stops short circuits.
- **Kapton** is thin amber heat-proof tape for holding cables down. It peels off and can be reused if clean.
- **Stretch-release tape** (under screens and batteries) lets go when you pull its tab slowly and flat. Pulling up snaps it.
- Conductive (shiny) tape on touchpads and cables must touch metal again when you refit, or the part acts strangely.

Skill 5: Static and battery safety



- Static from your body can kill a board without a spark you can see. Strap on, and touch boards only by the edges.
- A swollen, hot, hissing or sweet-smelling battery is an emergency: don't press it, move it to the fire-safe bin with an adult.
- Water, drinks and food stay off the bench.

Skill 6: Photograph, then test before you close



- Your photos are your instructions for putting it back together. Take more than you think you need.
- Plug in, power on and test the repaired part while the cover is still off. Finding a loose cable now saves taking it all apart again.
- Count your screws at the end. A leftover screw means something isn't fastened.

Word list

Word	What it means
Bezel	The plastic frame around the screen.
Bottom cover (base, D cover)	The panel underneath the Chromebook.
Top cover (palm rest, C cover)	The part your wrists rest on; often has the keyboard built in.
Lid / A cover	The back of the screen half.
Board / system board / mainboard	The main circuit board with the processor, memory and storage soldered on.
Daughterboard / I/O board	A small extra board holding some ports, joined to the main board by a ribbon cable.
Ribbon cable (FFC)	A flat, thin cable with gold stripes at the ends.
ZIF connector	A slot with a small flip-up latch that clamps a ribbon cable. "Zero insertion force": open it and the ribbon slides in or out freely.
eDP / LVDS cable	The video cable that carries the picture from the board to the screen.
Coax / antenna lead	The thin round wires from the Wi-Fi card to the antennas in the lid.
Spudger	A plastic or nylon pry stick. Safe to use near electronics.
Pick	A thin plastic triangle used to open seams and clips.
Captive screw	A screw that stays in the part after you loosen it.
Broad-head screw	A screw with a wide, flat head. Often used where it clamps plastic.
Mylar	Thin plastic insulating sheet.
Kapton tape	Thin amber heat-proof tape.
Thermal paste / pad	Grey paste or a soft pad that carries heat from the processor to the heat sink.
Heat sink	Copper or aluminium part that pulls heat off the processor.

Word	What it means
ESD	Electrostatic discharge: static electricity that can damage boards.
Isolator	A rubber piece that stops a part (usually a speaker) from rattling.
Powerwash	A factory reset of ChromeOS. Erases files saved on the device.
Recovery	Reinstalling ChromeOS from a USB stick or SD card.
EC reset (hard reset)	Restarting the small chip that controls power, charging and the keyboard.
Enroll	Connecting the Chromebook to the district's management so school policies apply.
eMMC	The storage chip on the board (like a small SSD). Soldered, so it can't be swapped.
Level 1 / 2 / 3	Who may do a repair: any trained student / student with sign-off / lead tech only.

2 Know the machine

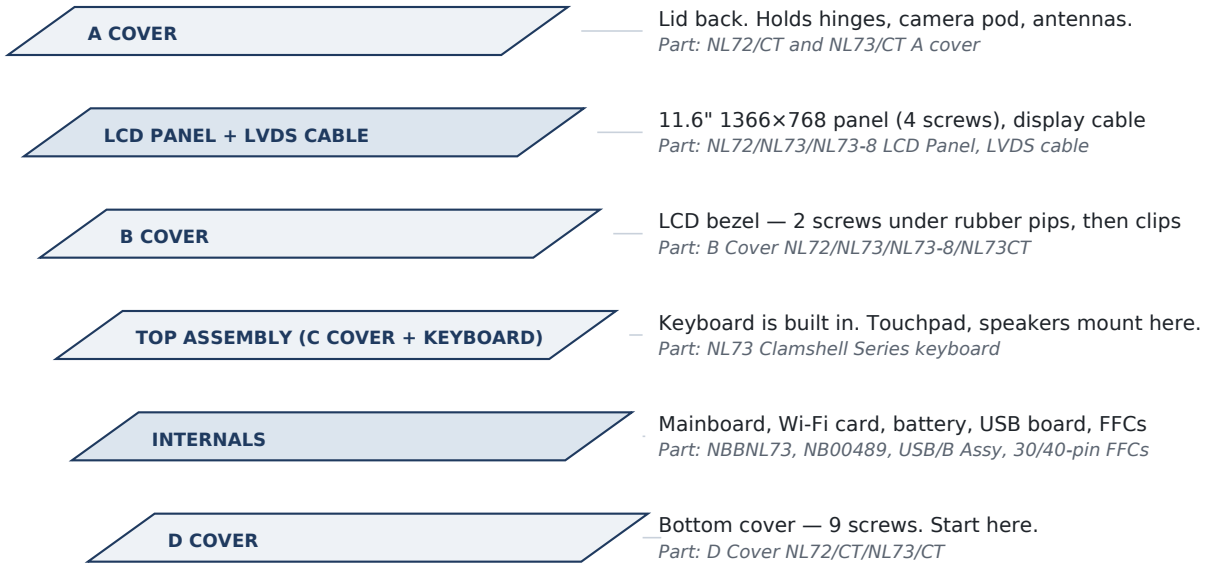
Specs that matter for repair

Item	NL73 / NL73-8	Why it matters
Display	11.6", 1366×768, non-touch (NL73CT: touch)	NL72/NL73/NL73-8 panel fits. NL73CT needs the touch panel.
Processor	Intel N100, quad-core	Soldered to mainboard
Memory	NL73: 4 GB • NL73-8: 8 GB LPDDR5	Soldered. Determines which mainboard to order.
Storage	64 GB eMMC	Soldered. Storage failure = mainboard.
Battery	Internal, one battery fits the whole NL73 series	NBBNL73. CTL doesn't publish its Wh rating.
Ports	2× USB-C, 2× USB-A, microSD, audio	One USB-C and the microSD slot sit on the USB board, per CTL's manual figures
Wireless	Wi-Fi 6E (Intel AX211), Bluetooth 5.3	Socketed card, 1 screw. Not the NL72's AX201.
Camera	720p HD, rotating camera pod	Module plus front cap, back cap, and hinge cap
Keyboard	Spill- and drop-resistant, anti-peel keys	Built into the top assembly. Single keys also sold.
Hinge	180° clamshell	Left and right sold separately
Durability	MIL-STD-810H, 75 cm drop, carry handle	
Weight	2.84 lb	
Updates	ChromeOS auto-updates to June 2033	
SKU	NL73 CBUS1100028 • NL73-8 CBUS1100024 • NL73CT CBUS1100026	Read the D-cover label

Specs from CTL's NL73 series spec sheet. The USB board's port assignments are inferred from the NL72 manual's figures. Confirm on teardown.

Anatomy: the A/B/C/D covers

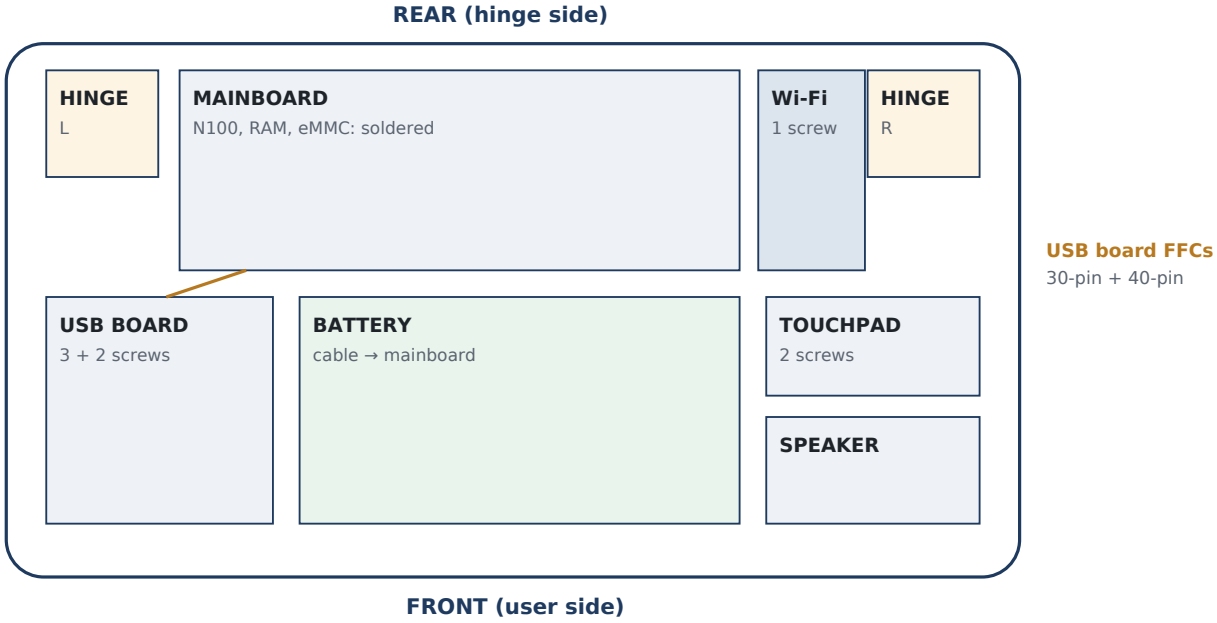
CTL names the shell pieces by letter, and parts are ordered by these names. On the NL73 the keyboard is part of the C cover, called the **top assembly**.



Top of stack = lid exterior. Bottom of stack = underside. Screw counts from CTL's NL72 manual, which CTL says to use for the NL73.

Inside the base

With the D cover off, you see the parts below. Wi-Fi sits at the right side of the mainboard, per CTL's manual.



ILLUSTRATIVE — screw counts are from CTL's NL72 manual; block positions are approximate.

NL73 parts list

Parts CTL lists as fitting the NL73 clamshell, as of October 2026. Stock changes often. Items marked † are NL72 listings that don't name the NL73. Ask CTL to confirm fit before ordering. Order at ctl.net/collections/chromebook-parts.

Part (CTL listing name)	SKU	Repair
Replacement Battery for NL73 Series (all NL73 models)	NBBNL73	R4
NL73 Clamshell Series keyboard (NL73, NL73-8, NL73CT)	NBKEYNL73	R17
US individual key replacement, NL73 series	—	R1
NL72, NL73 and NL73-8 LCD Panel (non-touch)	—	R11
NL72 and NL73 LVDS Cable (clamshell)	—	R12
B Cover NL72/NL73/NL73-8/NL73CT	—	R10
A cover NL72/CT and NL73/CT	—	R15
D Cover NL72/CT/NL73/CT	—	R3
D bottom cover screws, NL72 series†	—	R3
Left Hinge / Right Hinge NL72/CT/L/NL73/CT	—	R14
Camera Module NL72/NL72L/NL73/NL73-8/NL73CT	NB00430	R13
Front Camera Cap, Back Camera Cap, Camera Hinge (cap)	—	R2
USB/B Assy for NL73 Clamshell Series (listed as "USD/B")	—	R8
30-pin and 40-pin USB/B ribbon cables	—	R9
Touch Pad, NL72 series†	—	R7
Left / Right Speaker, NL72†	—	R16
Mainboard NL73 4/64	NB00489	R19
Mainboard NL73-8 and NL73CT (8/64)	—	R19
CTL 45 W USB-C Adapter	—	Triage

Don't use the NL72's Wi-Fi card. CTL sells an AX201 Wi-Fi module for the NL72. The NL73 ships with an AX211 (Wi-Fi 6E). Ask CTL for the correct NL73 card rather than fitting the NL72 part.

Stocking suggestion. Keep camera caps, individual keys, D-cover screws, and both USB ribbons on hand. They're small, cheap, and cover the most common tickets.

3 Triage: symptom → repair

Start with the cheapest, least invasive fix. Work through the "try first" column before ordering parts.

Symptom	Try first	Then	Part likely needed
Won't turn on, no lights	Known-good charger 30 min → S1	R5 reseat battery → R4	Battery, or mainboard
Not charging	Other USB-C port, other charger → S1	S2 battery test → R4 → R8 if one port only	Battery, USB board, or mainboard
Battery drains fast / low health	S2 battery test	R4	Battery
Swollen battery / bulging base	STOP — Level 3	R4 by lead tech	Battery
Black screen, power light on	External display via USB-C; S1	R5 → R12 → R11	LVDS cable or panel
Flicker or lines that change with lid angle	S1	R12 LVDS cable	LVDS cable
Cracked or blotchy screen	—	R11 LCD panel	LCD panel
Bezel loose or broken	Reseat clips, check 2 screws	R10	B cover
Lid cracked	—	R15	A cover
Lid floppy, hinge cracked	Check hinge screws	R14	Hinge(s)
Camera cap missing, camera pod loose	—	R2	Camera caps
Camera not detected	S1, check camera pod rotation	R13	Camera module
One key missing or broken	—	R1	Individual key
Several keys or a row dead	S2 keyboard test; S1	R5 reseat keyboard FPC → R17	Top assembly
Touchpad dead or erratic	S1; check touchpad settings	R5 → R7	Touchpad or its FFC
USB-A, microSD, or one USB-C port dead	Test with another device; S1	R9 ribbons → R8 USB board	Ribbons or USB board
No sound	Volume, output setting; S1; S3	R5 speaker cable → R16	Speakers
Wi-Fi missing or weak	S1 → S3	R6 reseat card and antennas	Wi-Fi card (ask CTL)
Liquid spill	Power off NOW, unplug	R18 (Level 3)	Depends on damage
Slow, crashing, errors	S1 → S3	Sec. 6 if recovery fails	Often none
"ChromeOS is missing or damaged"	S3 recovery	R19 if recovery can't write	Mainboard
Base cover cracked or screws missing	—	R3	D cover or screws

4 Software repairs

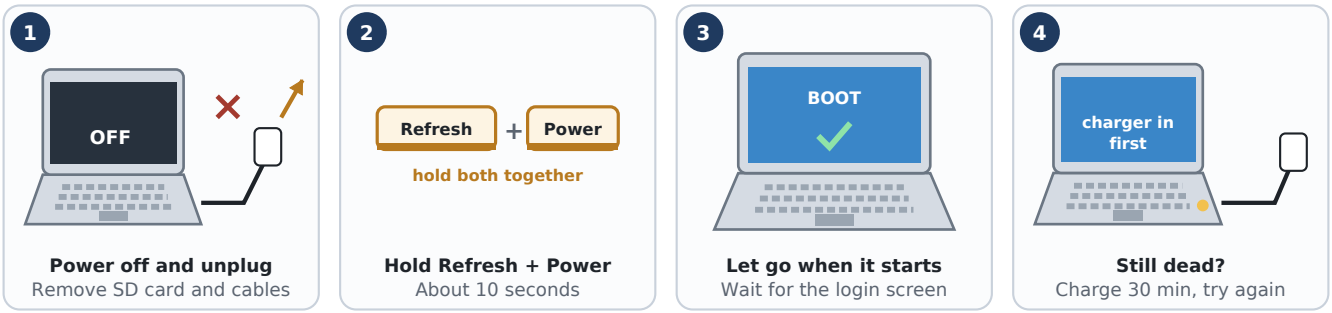
These need no tools and clear many tickets. Always do S1 before opening a device.

S1 Hard reset (EC reset) LEVEL 1

Fixes	Won't power on, won't charge, frozen, black screen, unresponsive keyboard or touchpad
Time	2 minutes
Tools	Charger
Data loss	None

In plain words. A hard reset restarts the small chip that runs power, charging and the keyboard. It's like unplugging something and plugging it back in, and it fixes a surprising number of "dead" Chromebooks. Nothing is erased.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

CTL's manual says to try a power cycle before any reset. A hard reset restarts the embedded controller, which manages power, keyboard, and charging.

Step by step

1. ☐ Unplug the charger and any USB devices. Remove the microSD card.
2. ☐ Hold Power for 10 seconds so the device is fully off.
3. ☐ Hold **Refresh** (circular arrow), then tap **Power**. Release Refresh when it starts.
4. ☐ If nothing happens, plug in a known-good 45 W USB-C charger, wait 30 minutes, then repeat steps 2–3 with the charger connected. Try the other USB-C port too.

Common mistakes
- ✗ Giving up after one try without the charger.
- ✗ Holding only Power (that's just a shutdown).
- ✗ Using a charger you haven't tested.

If something goes wrong
Nothing at all, even charging → Try the other USB-C port, then check the battery (battery repair).
Starts then shuts off → Run diagnostics (S2) for the battery.

Verify before closing the ticket

- Device reaches the login screen.
- Charging light comes on with the charger connected.

Tip. After the battery has been disconnected (R4, R5), plug in a charger before deciding a device is dead. Many Chromebooks won't wake on the power button alone.

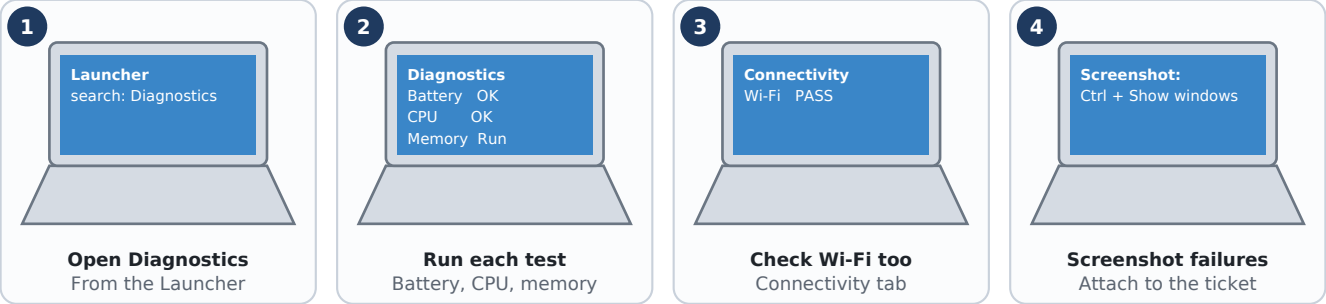
S2 Built-in hardware diagnostics

LEVEL 1

Fixes	Confirms the fault before you order a part: battery, CPU, memory, keyboard, touchpad, network
Time	5–15 minutes
Tools	None
Data loss	None

In plain words. Chromebooks have a built-in checkup app. It tests the battery, processor, memory and more, and tells you what's failing before you order anything.

At a glance



Open Diagnostics
From the Launcher

Run each test
Battery, CPU, memory

Check Wi-Fi too
Connectivity tab

Screenshot failures
Attach to the ticket

The pictures show the idea, not your exact model. The numbered steps below have the details.

Step by step

1. ☐ Sign in with a bench test account if you can.
2. ☐ Open the **Launcher**, search **Diagnostics**, and open it.
3. ☐ **System:** read battery health and cycle count, then run the battery charge and discharge tests.
4. ☐ Run the CPU and memory tests.
5. ☐ **Connectivity:** confirm Wi-Fi is detected and run the tests.
6. ☐ **Input:** open the keyboard tester and press every key. Test the touchpad.
7. ☐ Screenshot any failed result (Ctrl + Show windows) and attach it to the ticket.

Common mistakes
✗ Skipping the memory test because it's slow.
✗ Not saving the results to the ticket.
✗ Replacing a battery that tests healthy.

If something goes wrong
Battery health low → Battery repair.
Memory or storage fails → System board: Section 6 or lead tech.

Verify before closing the ticket

- Results recorded on the ticket.
- Parts ordered only for a failed test or a fault you saw yourself.

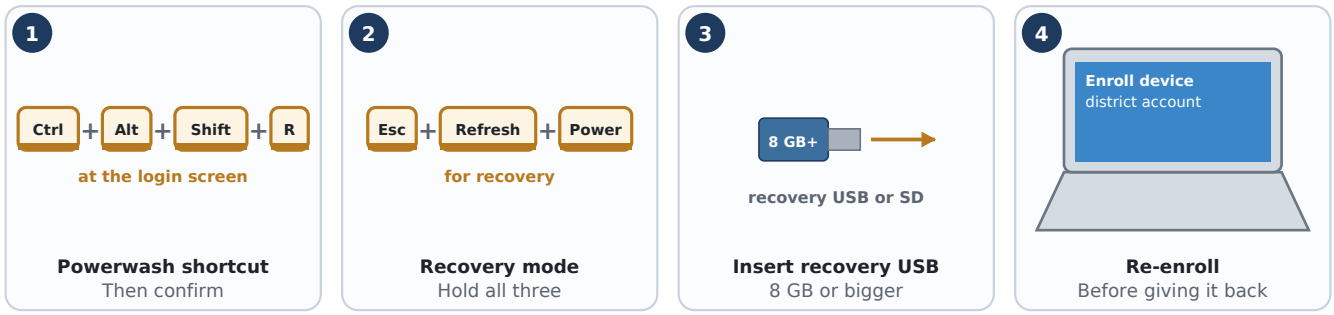
S3 Powerwash and USB recovery

LEVEL 1

Fixes	Slow, crashing, corrupted OS, "ChromeOS is missing or damaged", outdated OS
Time	Powerwash 10 min; recovery 20–40 min
Tools	Recovery USB or SD card (8 GB+)
Data loss	**Yes.** Recovery permanently erases all local storage.

In plain words. A Powerwash wipes the Chromebook back to factory settings, like a fresh start. Recovery goes further and reinstalls ChromeOS from a USB stick. Both erase files saved on the device, so warn the student first.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

District-managed devices. A powerwash or recovery on an enrolled device should force it to re-enroll on its first Wi-Fi connection. Confirm with the lead tech that forced re-enrollment is on and the bench Wi-Fi can reach enrollment. Never enter Developer Mode on a district device.

Step by step

1. ☐ **Check for updates first:** Settings > About ChromeOS > Check for updates. An outdated OS causes many odd faults.
2. ☐ **Powerwash (factory reset):** sign out, press Ctrl + Alt + Shift + R, select Restart, then Powerwash > Continue. (May be blocked by policy on managed devices. If so, use recovery.)
3. ☐ **Make recovery media:** on a working computer, install the Chromebook Recovery Utility, open it, and find the CTL NL73 by searching its model name or picking from the list. Insert an 8 GB+ USB drive or SD card and click Create now.
4. ☐ On the NL73, hold **Esc + Refresh**, tap **Power**, and release the keys when the recovery message appears.
5. ☐ Insert the recovery media and follow the prompts.
6. ☐ After it reboots, connect to Wi-Fi and confirm the device re-enrolls to the KPS domain.

Common mistakes

✗ Not warning that Downloads will be erased.
✗ Choosing Developer Mode (never on district devices).
✗ Handing it back before re-enrolling.

If something goes wrong

"ChromeOS is missing or damaged" again after recovery → Storage failing: system board, Section 6.
Can't enroll → Lead tech checks the Admin console.

Verify before closing the ticket

- Device boots to the enrollment or KPS sign-in screen.
- Shows as enrolled in the Admin console.
- If recovery can't write to storage, the eMMC has failed: R19 or Sec. 6.

Recovery error messages, from CTL's manual: **"An unexpected error has occurred"** — erase the media and retry, or use a different drive. **"Please remove all connected devices"** — unplug everything and restart recovery. **"Does not contain Chrome OS"** — wrong model image or bad drive. **"Security module not working"** — escalate to the lead tech. The NL72 manual's recovery model code doesn't apply to the NL73: choose the NL73 in the utility, or ask CTL for the image.

5 Hardware repairs

Most hardware repairs start with R3 (base cover off) and the first steps of R4 (battery disconnected). They're written out once and referenced everywhere else. Screw counts in **bold** come from CTL's NL72 manual.

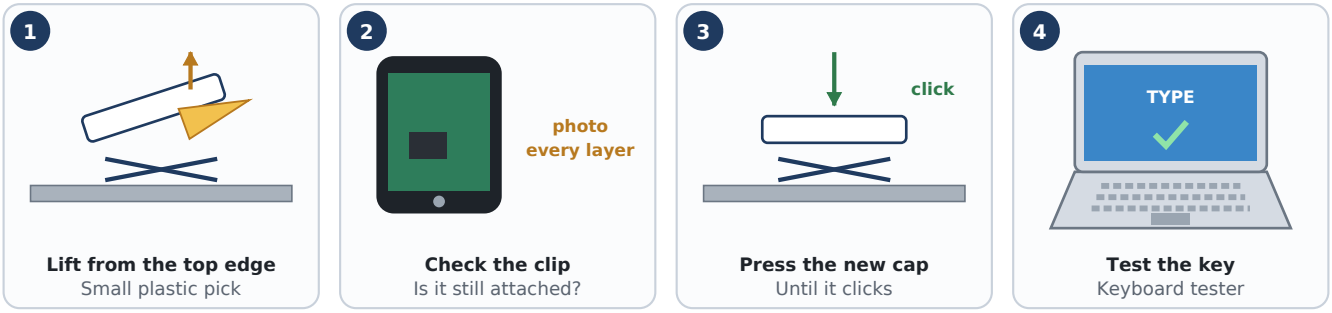
Golden rules. Photograph every layer before removing it. Lay screws on the mat in the same pattern as the device. Unlock every connector before pulling a cable. Never force a part: if it won't move, find the screw or clip you missed. Test before the last screws go in.

R1 Individual key replacement — LEVEL 1

Fixes	Missing, cracked, or loose key
Part	US individual key replacement for the NL73 series
Time	5 minutes
Tools	Plastic pick, tweezers

In plain words. A key is a plastic cap clipped onto a small scissor-shaped hinge. A missing or broken key can usually be fixed with just the cap, in a few minutes.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model

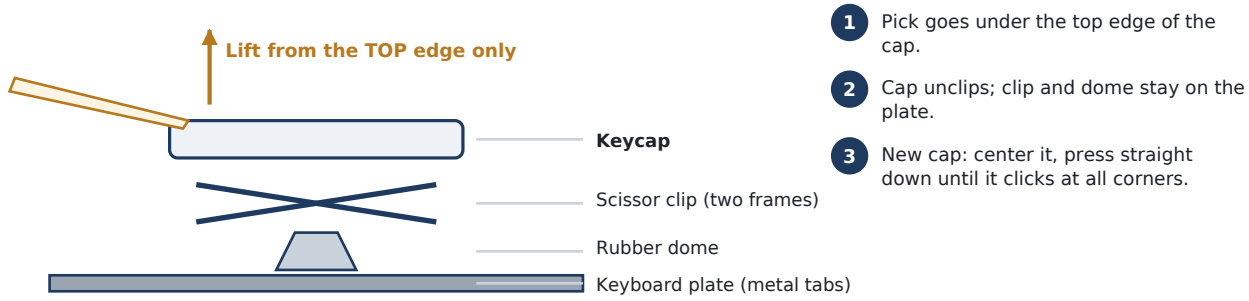


Illustration: side view of one key. Not to scale. The NL73 has anti-peel keys: go gently.

The NL73 has anti-peel keys, built to resist being pried off. Work gently and lift from the top edge only.

Step by step

1. ☐ Power off. Open the lid flat on the mat.
2. ☐ Slide a plastic pick under the top edge of the damaged key and lift until it unclips.
3. ☐ Inspect the scissor clip and rubber dome. If the clip is broken, ask CTL whether the replacement key includes one.
4. ☐ To fit a clip: hook its pins under the metal tabs on the keyboard plate, then press it flat.
5. ☐ Center the new key over the clip and press straight down until it clicks at every corner.
6. ☐ Wide keys (Space, Shift, Enter, Backspace): seat the metal stabilizer wire in its holders first.

Common mistakes
✗ Prying from the bottom edge (snaps the clip).
✗ Pressing a cap on crooked.
✗ Using a cap from a different model.

If something goes wrong
Clip is broken → Order the key with its clip, or replace the keyboard.
Key types nothing → Keyboard ribbon or keyboard: connectors repair first.

Verify before closing the ticket

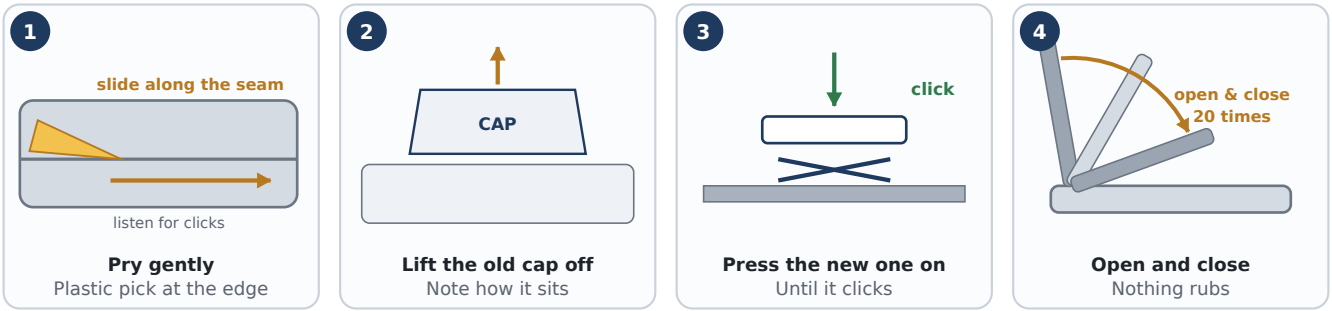
- Key clicks evenly and springs back.
- Key registers in the S2 keyboard tester.

R2 Camera caps and camera hinge cap — LEVEL 1

Fixes	Missing, cracked, or loose plastic around the rotating camera
Part	Front Camera Cap, Back Camera Cap, or Camera Hinge (cap)
Time	10–15 minutes
Tools	Plastic picks

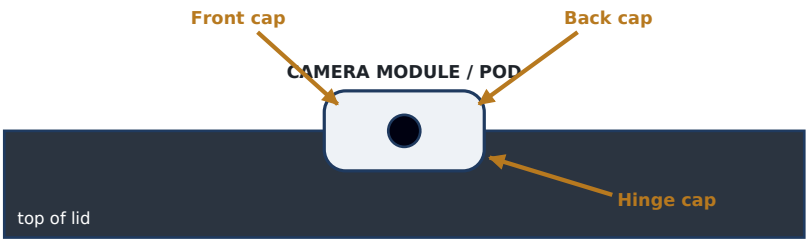
In plain words. These are small plastic covers that snap on. They mostly protect and tidy; replacing one is a quick clip-off, clip-on job.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Bezel off first (it covers the camera's front).
2. Unplug the small camera cable with tweezers.
3. Lift the module by its board edges, never the lens.
4. Test in the Camera app before closing up.

Illustration: camera at the top-center of the lid. Cap shapes simplified.

The NL73's camera sits in a pod at the top of the lid. Three small plastic parts cover it, and students bump and knock them off often. Some may require the bezel off (R10). Confirm on your first one and record it below.

Step by step

1. ☐ Power off and unplug.
2. ☐ Look closely at which cap is damaged: front (screen side), back (lid side), or the hinge cap the pod turns on.
3. ☐ Pry the damaged cap off with a plastic pick, starting at a seam. If it won't release, remove the bezel (the plastic frame around the screen) (R10) and check for screws or tabs from behind.
4. ☐ Check the camera module and its cable for damage. If either is damaged, do R13 now.
5. ☐ Line up the new cap and press it until every clip seats.

Common mistakes
✗ Using a metal tool.
✗ Forcing a cap the wrong way round.

If something goes wrong
Cap won't stay on → Clip broken: order a new cap.

Verify before closing the ticket

- Cap is flush with no gap.
- Camera pod rotates fully and holds position.
- Camera app shows a clear image both ways.

First-teardown notes — fill in the first time you do this repair	
Front cap: clips only, or bezel off?	_____
Back cap: clips only, or bezel off?	_____
Camera hinge cap: how removed?	_____

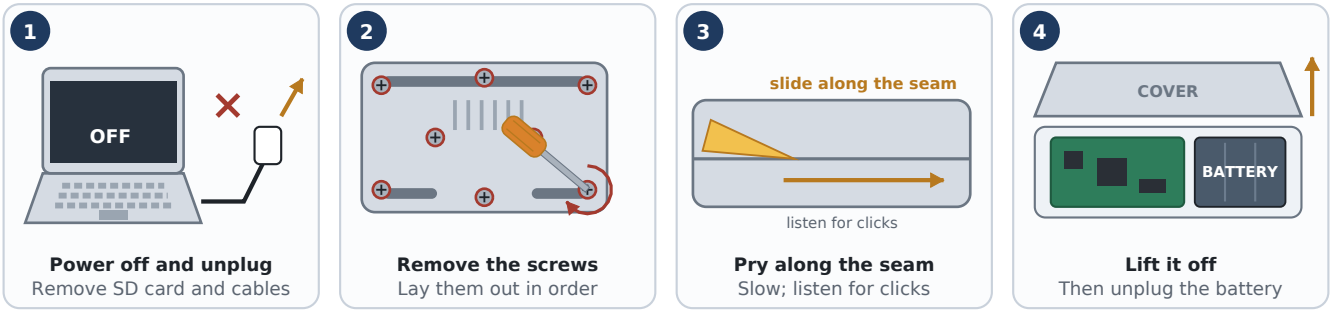
R3 Base cover (D cover) removal and replacement

LEVEL 1

Fixes	Cracked or missing D cover, missing screws. Also the first step for R4–R19.
Part	D Cover NL72/CT/NL73/CT • D-cover screws (sold individually)
Time	10 minutes
Tools	Phillips #0 or #1, plastic pry tool, screw mat

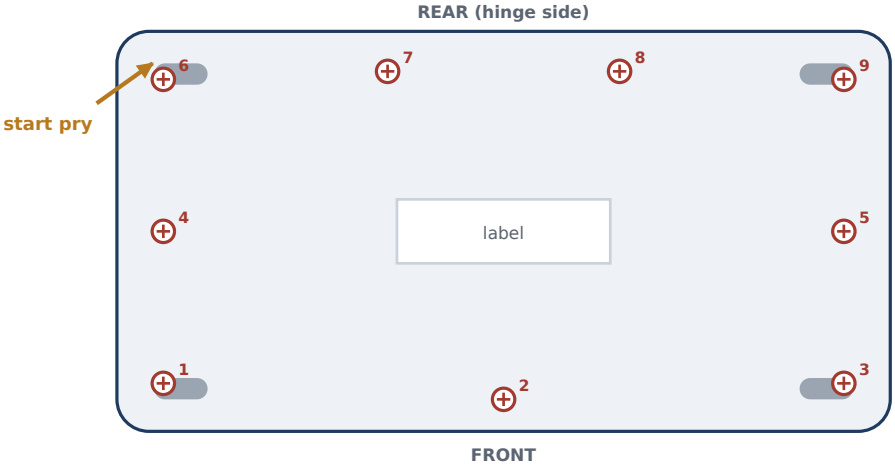
In plain words. The bottom cover is the plastic panel underneath. It's held by small screws and plastic clips. Taking it off is the first step for almost every repair inside.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Remove all 9 screws (CTL count).
2. Lay screws on the mat in this same pattern.
3. Pry from the corner shown, then work around the edge.
4. Disconnect the battery before anything else.

Illustration: underside. 9 screws per CTL's manual; exact positions illustrative.

Step by step

1. ☐ **Pre-disassembly (from CTL's manual):** place the device on a stable surface. Remove the charger, the microSD card, and all cables.
2. ☐ Close the lid and flip the device over onto the mat.
3. ☐ Remove the **nine (9)** base-cover screws. Place each on the screw mat in its position.
4. ☐ Starting at a corner, pry the cover open with your thumb and forefinger, a pry tool, or a spudger (a plastic pry stick).

5. ☐ Work around the edge to release the clips, then lift the cover off.
6. ☐ **Stop and disconnect the battery (R4 steps 2–3) before anything else.**
7. ☐ To refit: line up the cover, press around the edge until every clip snaps, then refit all 9 screws. Snug only.

Common mistakes
✗ Missing a screw under a rubber foot and cracking the cover.
✗ Mixing up long and short screws.
✗ Using a metal tool in the seam (scratches, cracks).

If something goes wrong
Cover won't release in one spot → There's still a screw there. Check under the feet and labels.
A clip broke → It usually still holds. If it rattles, order a new cover.

Verify before closing the ticket

- No gaps around the seam.
- All 9 screws refitted. Device sits flat.

First-teardown notes — fill in the first time you do this repair

Screwdriver size that fits	_____
Screw lengths (all the same?)	_____
Hidden screws under feet? (Y/N)	_____

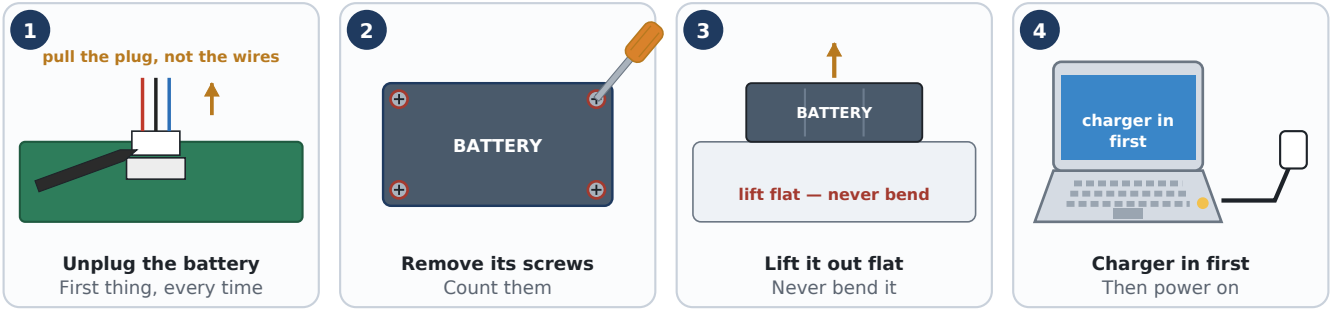
R4 Battery replacement

LEVEL 2

Fixes	Won't hold charge, low health in S2, swollen battery
Part	Replacement Battery for NL73 Series, NBBNL73
Time	15–20 minutes
Tools	Phillips driver, plastic spudger, ESD strap

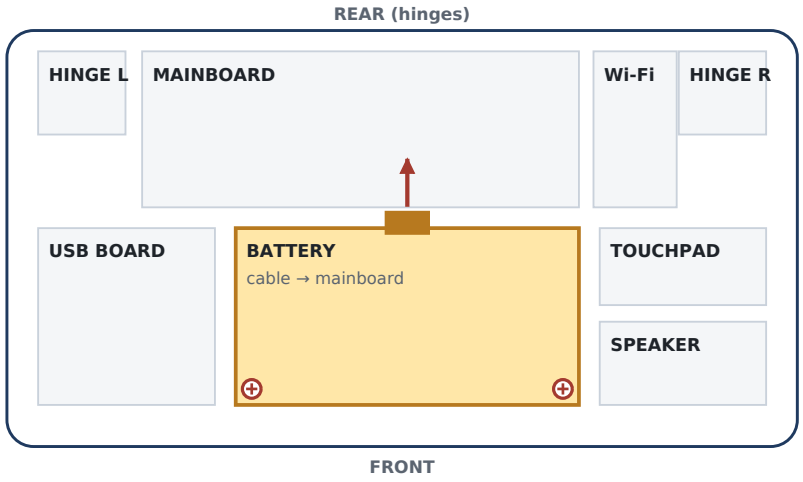
In plain words. The battery is a flat pack that powers the Chromebook when it's unplugged. To replace it you unplug its cable from the board, remove a few screws, and lift it out. Batteries can be dangerous if bent or punctured, so handle them gently.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Unplug the battery cable from the mainboard first (red arrow).
2. Pry at the plug body, never the wires.
3. Remove the battery screws (count them).
4. Lift out flat. Never bend or pry against the cell.

Illustration: base interior, battery highlighted. Screw positions illustrative.

Swollen battery = Level 3. Don't press or pry. The lead tech handles it, and the old cell goes straight to the fire-safe bin.

Step by step

1. ☐ Do R3 to remove the base cover.
2. ☐ Find the battery and its cable. Per CTL's manual, the cable runs from the battery to a connector on the mainboard.
3. ☐ Disconnect the battery cable from the mainboard connector. Pry at the connector body, never the wires.
4. ☐ Remove the screws holding the battery. CTL's manual doesn't list how many. Record it below.
5. ☐ Lift out the battery. Never bend it.
6. ☐ Fit the new battery, refit its screws, and reconnect the cable firmly.
7. ☐ Refit the D cover (R3 step 7).
8. ☐ Plug in the charger before powering on.

Common mistakes

✗ Pulling the battery cable by its wires.
✗ Prying under the battery with a metal tool.
✗ Thinking it's dead because it won't start without the charger.

If something goes wrong

Swollen, hot, hissing or smells sweet → STOP. Don't press it. Fire-safe bin, get an adult.
Won't power on with new battery → Plug in the charger for 30 minutes, then hard reset (S1).

Verify before closing the ticket

- Charging light comes on.
- Device powers on and shows a battery percentage.
- S2 shows near-100% health on the new battery.
- Old battery goes to recycling. CTL offers free battery recycling through Call2Recycle.

First-teardown notes — fill in the first time you do this repair	
Number of battery screws	_____
Connector type: lift-up or slide-out?	_____
Adhesive under battery? (Y/N)	_____

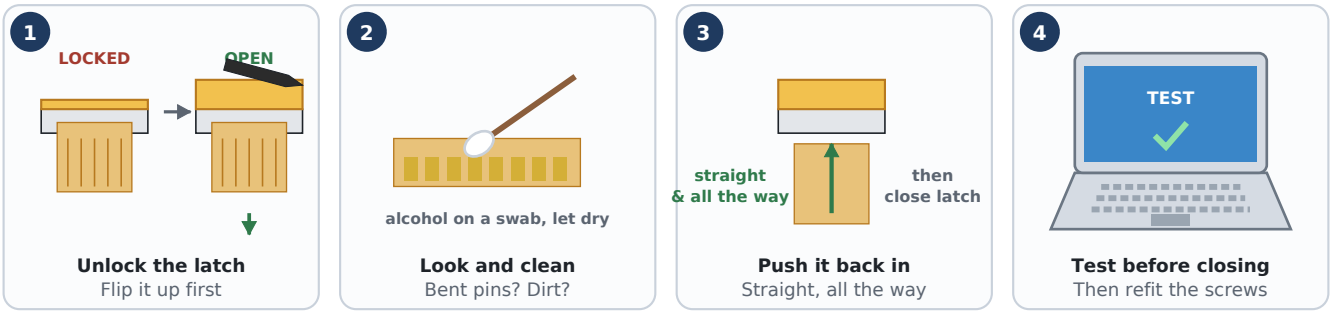
R5 Reseating internal connectors

LEVEL 2

Fixes	Dead keyboard, touchpad, or speakers; black screen; dead ports; no power after a drop
Part	None, unless a cable is damaged
Time	15–25 minutes
Tools	Plastic spudger, plastic tweezers, isopropyl alcohol

In plain words. Inside, parts connect to the main board with small plugs and flat ribbon cables. A drop can wiggle one loose. Unplugging it, checking it, and plugging it back in properly fixes many problems for free.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model

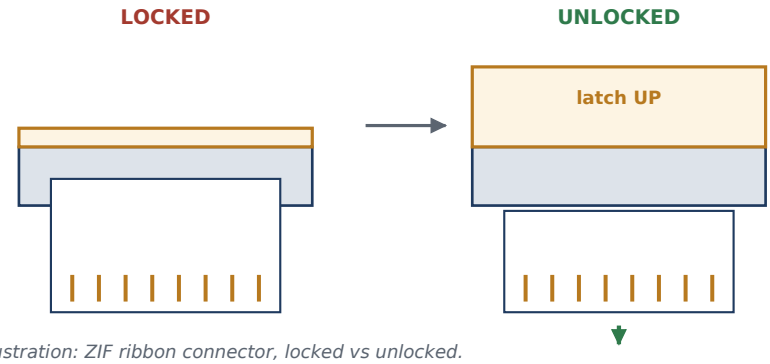


Illustration: ZIF ribbon connector, locked vs unlocked.

1. Flip the hinged latch up (or slide a drawer-type latch out) with a spudger.
2. Ribbon now slides out with zero force. Never pull a locked ribbon.
3. Refit: contacts facing the same way, push in straight and fully, close latch.
4. Board-to-wire plugs (battery, speaker): pry at the plug body, never the wires.

Connector (from CTL's manual)	Type	How to release
Battery cable	Board-to-wire	Pry at the connector body
LVDS (display) cable	Board-to-wire with locking arm	Remove the 2-screw protective plate, lift the locking arm, then pull

Connector (from CTL's manual)	Type	How to release
Touchpad FFC	ZIF ribbon	Unlock the latch first or the cable is damaged
Keyboard FPC	ZIF ribbon	Unlock the latch first
USB board FFCs (30-pin, 40-pin)	ZIF ribbon	Unlock both ends first
Speaker cable	Board-to-wire	Pry at the connector body
Wi-Fi antennas (2)	Snap-on coax	Lift straight up with tweezers

Step by step

1. ☐ Do R3 and disconnect the battery.
2. ☐ Hold Power for 5 seconds with the battery disconnected to drain leftover power.
3. ☐ For each connector related to the symptom: photograph it, unlock and release it, and inspect it for bent pins, burn marks, or creases.
4. ☐ Clean dirty or corroded contacts with a brush dampened in isopropyl alcohol. Let them dry for 5 minutes.
5. ☐ Reconnect each one. Ribbons must go in straight and all the way before you close the latch. A crooked ribbon is the #1 cause of "still broken."
6. ☐ Reconnect the battery last. Plug in and test **before** refitting the D cover.

Common mistakes

✗ Pulling a ribbon without unlocking the latch.
✗ Inserting a ribbon crooked or only half way.
✗ Forgetting to reconnect the battery last.

If something goes wrong

Ribbon is creased or torn → Replace that cable or part.
Symptom comes back later → The part itself is failing: replace it.

Verify before closing the ticket

- Symptom gone in a full test, not just at boot.
- Creased or torn ribbons replaced (R9, R12).

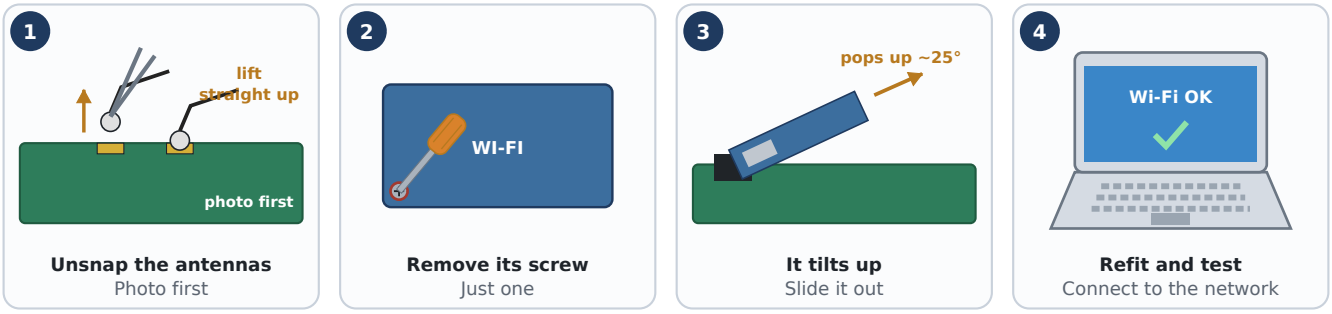
R6 Wi-Fi card reseat or replacement

LEVEL 2

Fixes	No Wi-Fi or Bluetooth, weak signal, Wi-Fi drops after a drop
Part	Usually none (reseat). Replacement: ask CTL for the NL73's AX211 card. Don't use the NL72's AX201.
Time	15 minutes
Tools	Phillips #0, plastic tweezers

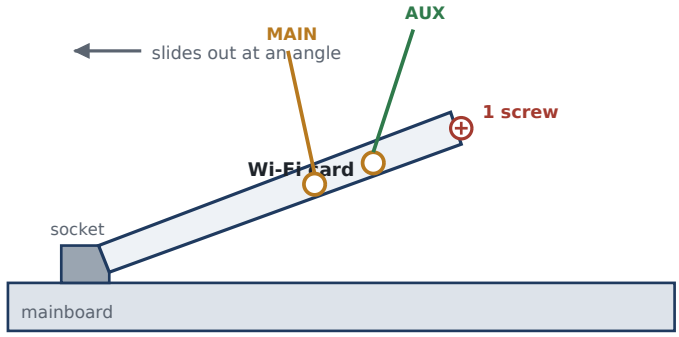
In plain words. The Wi-Fi card is a small removable card with two thin antenna wires snapped on top. It sits at an angle in a slot and is held down by one screw.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Photograph which antenna goes on which terminal.
2. Lift both antenna plugs straight up with plastic tweezers.
3. Remove the 1 screw. The card springs up at an angle.
4. Slide it out of the socket. Refit in reverse.

Illustration: M.2 Wi-Fi card, side view. Colors of antenna leads vary.

CTL's manual lists the WLAN module as self-repairable. It's a socketed card held by one screw, with two antenna leads.

Step by step

1. ☐ Do R3 and disconnect the battery.
2. ☐ Find the Wi-Fi card at the right side of the mainboard.
3. ☐ Note which antenna lead goes on which terminal (photo). Lift both leads straight off with plastic tweezers.
4. ☐ Remove **one (1)** screw.
5. ☐ The card tilts up. Slide it out of the mainboard socket.

6. ☐ To reseat or replace: slide the card into the socket at an angle, press it down, refit the screw, and snap both antennas on in the same positions.
7. ☐ Reconnect the battery and test.

Common mistakes
✗ Swapping the two antenna wires.
✗ Pulling antennas sideways (breaks the snap).
✗ Using a card that isn't the approved part.

If something goes wrong
Weak signal after a repair → An antenna wire is pinched or unplugged.
No Wi-Fi at all → Try a powerwash (S3) and a known-good card.

Verify before closing the ticket

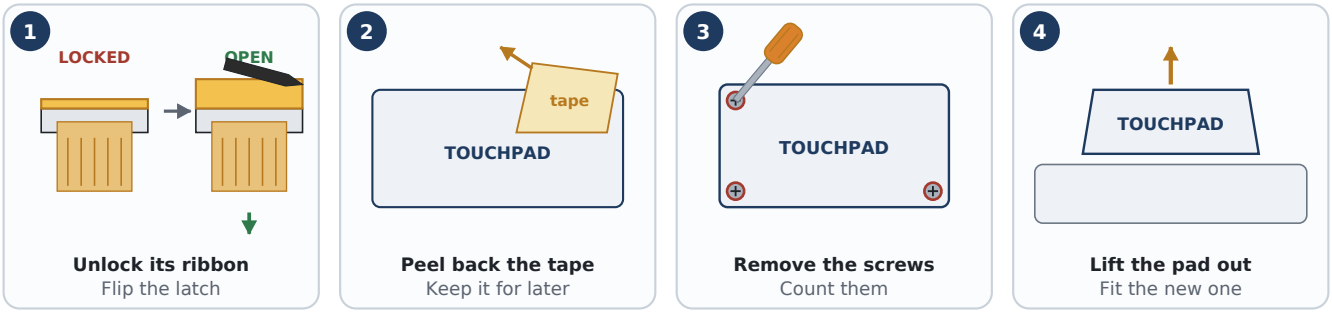
- Wi-Fi connects and S2 connectivity tests pass.
- Bluetooth finds a device.
- Weak signal after a lid repair usually means an antenna lead got pinched in the hinge. Check the routing.

R7 Touchpad replacement — LEVEL 2

Fixes	Touchpad dead, erratic, or physically damaged, with the keyboard working
Part	Replacement Touch Pad for CTL NL72 Series (confirm NL73 fit with CTL)
Time	25 minutes
Tools	Phillips #0, plastic tweezers, Kapton tape

In plain words. The touchpad is a flat glass or plastic pad held from underneath by a few small screws and a bracket. A ribbon cable connects it to the board.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model

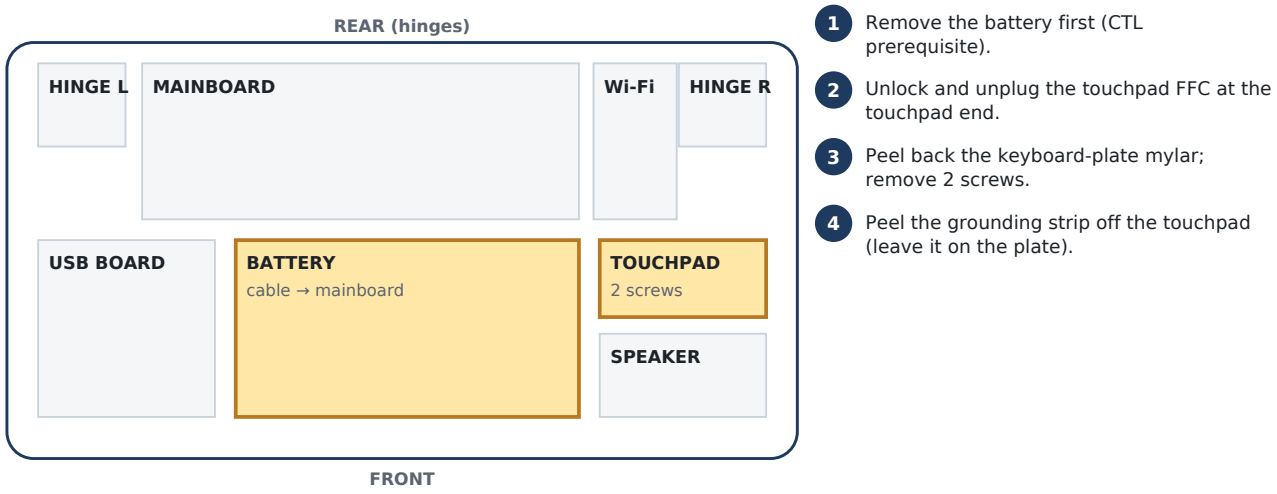


Illustration: base interior. Touchpad sits under the battery area; position illustrative.

Check first: touchpad enabled in Settings, S1 done, and R5 reseat tried.

Step by step

1. ☐ Do R3 and remove the battery (R4 steps 2–5). CTL lists battery removal as a prerequisite.
2. ☐ Find the touchpad module and its FFC (flat ribbon cable).
3. ☐ Unlock the latch, then disconnect the touchpad FFC from the touchpad connector. Leave the other end on the mainboard. **Don't pull a locked FFC.**
4. ☐ Peel back the keyboard plate mylar. Keep it intact to restick.
5. ☐ Remove **two (2)** screws. Disconnect the remaining cable at the touchpad connector. Peel back the grounding strip, leaving it attached to the metal plate.
6. ☐ Remove the touchpad module.
7. ☐ Fit the new touchpad, refit 2 screws, restick the grounding strip and mylar, and reconnect the FFC and latch.
8. ☐ Refit the battery, reconnect, and test before closing.

Common mistakes
✗ Forgetting the conductive tape (the pad acts strange).
✗ Over-tightening screws (click stops working).
✗ Ribbon not locked.

If something goes wrong
Click doesn't work → Loosen the bracket screws a little.
Cursor jumps around → Check the tape and ribbon; try S3.

Verify before closing the ticket

- Cursor moves, clicks, and two-finger scrolls.
- Grounding strip reattached. Without it, the touchpad can behave erratically.

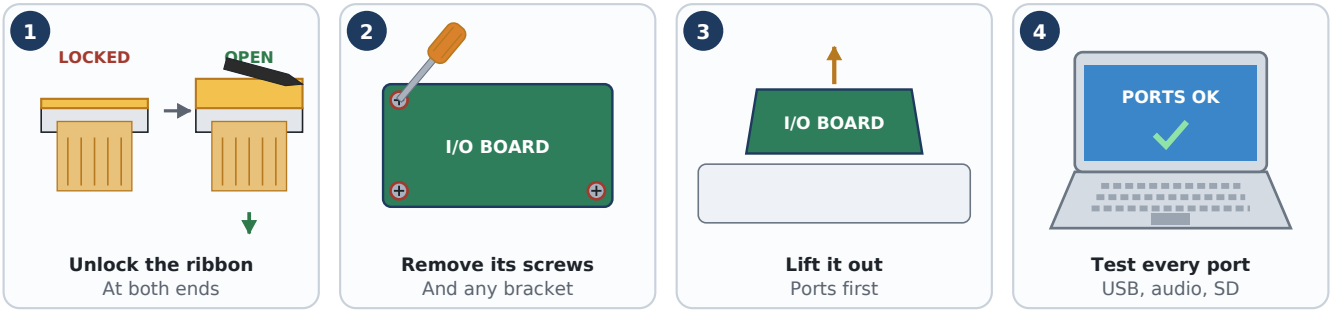
R8 USB board replacement

LEVEL 2

Fixes	Dead USB port or microSD slot on the USB board side; one USB-C port won't charge
Part	USB/B Assy for the NL73 Clamshell Series
Time	25–40 minutes
Tools	Phillips #0/#1, plastic tweezers

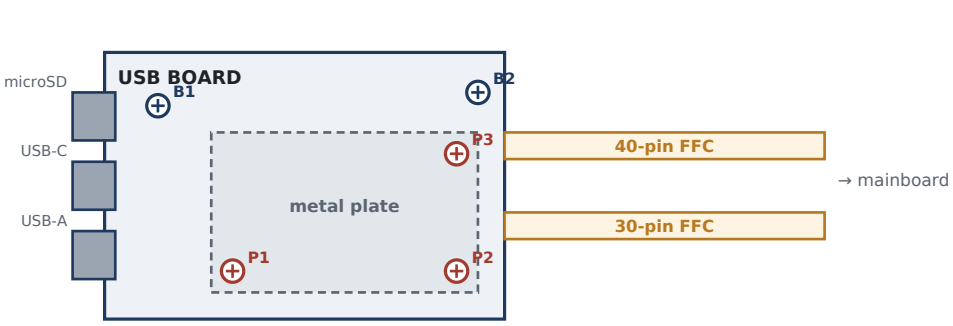
In plain words. The I/O board is a small separate board that holds some of the ports (like the USB-A port or headphone jack). It connects to the main board with a ribbon cable, so it can be swapped by itself.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Remove the 3 plate screws (P), then the plate.
2. Remove the 2 board screws (B).
3. Unlock both connectors, then slide out the 30- and 40-pin ribbons.
4. Lift the board, tilting ports out of the case.

Illustration: USB board from above. Port order and screw spots illustrative. Confirm on yours.

Try R9 first: a ribbon is cheaper than the board. CTL's manual lists the LCD module as a prerequisite, so the lid may need to come off to reach this board. Check on your first teardown whether you can reach the screws with the lid attached.

Step by step

1. ☐ Do R3 and disconnect the battery.
2. ☐ If the board's screws or plate are under the hinge area, remove the top half of the unit (R14 steps 2–5).
3. ☐ Find the USB board.
4. ☐ Remove **three (3)** screws from the protection plate, then remove the plate.

5. ☐ Remove the remaining **two (2)** screws on the board.
6. ☐ Unlock both connectors, then disconnect the 30-pin and 40-pin FFCs.
7. ☐ Remove the board. Fit the new one in reverse: 2 board screws, both FFCs locked, plate with 3 screws.
8. ☐ Reconnect the battery and test every port before closing.

Common mistakes
✗ Ribbon in upside down or crooked.
✗ Forgetting the bracket.
✗ Testing only one port.

If something goes wrong
Ports still dead → Ribbon cable or the main board connector.
USB-C dead → Usually on the main board: Section 6.

Verify before closing the ticket

- Every port on the board works: USB-A reads a drive, USB-C charges, microSD reads.
- Plate refitted with all 3 screws.

First-teardown notes — fill in the first time you do this repair

Which ports are on the USB board? _____

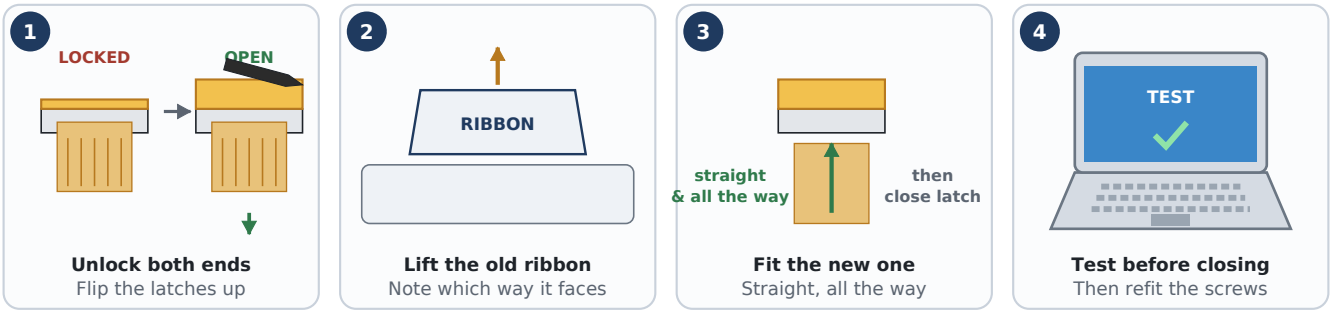
Did the lid have to come off? (Y/N) _____

R9 USB board ribbon cables (30-pin and 40-pin) LEVEL 2

Fixes	Ports on the USB board dead or intermittent, especially after a drop
Part	30-pin USB/B ribbon • 40-pin USB/B ribbon
Time	15–20 minutes
Tools	Plastic tweezers, Kapton tape

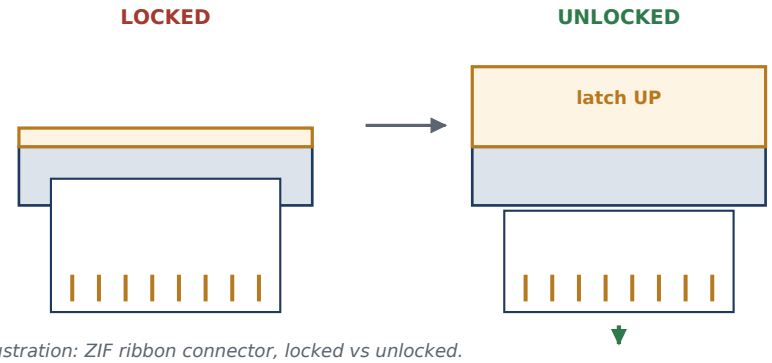
In plain words. These flat ribbon cables connect the small port board to the main board. A creased or loose ribbon kills the ports even when both boards are fine.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Flip the hinged latch up (or slide a drawer-type latch out) with a spudger.
2. Ribbon now slides out with zero force. Never pull a locked ribbon.
3. Refit: contacts facing the same way, push in straight and fully, close latch.
4. Board-to-wire plugs (battery, speaker): pry at the plug body, never the wires.

Illustration: ZIF ribbon connector, locked vs unlocked.

Step by step

1. ☐ Do R3 and disconnect the battery.
2. ☐ Photograph each ribbon's route and tape.
3. ☐ Unlock the connectors at both ends (USB board and mainboard), then slide the ribbon out.
4. ☐ Match the replacement: 30-pin or 40-pin.
5. ☐ Slide the new ribbon in at both ends, contacts the same way as the old one, and lock both latches.
6. ☐ Re-tape and test before closing.

Common mistakes

✗ Fitting the ribbon upside down.
✗ Creasing it while routing.
✗ Closing a latch on a half-inserted ribbon.

If something goes wrong

Ports still dead → Try the I/O board itself, or the board's connector: lead tech.

Verify before closing the ticket

- All USB-board ports work.
- Ribbons lie flat without a kink.

Replace both ribbons at once. They're cheap, and it saves reopening the device.

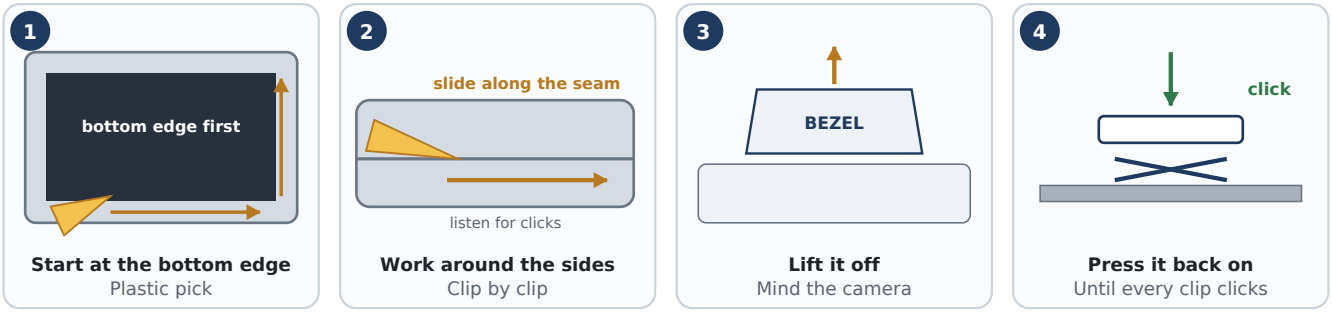
R10 LCD bezel (B cover) replacement

LEVEL 2

Fixes	Cracked, loose, or missing bezel. First step for R11, R12, R13.
Part	B Cover for NL72/NL73/NL73-8/NL73CT
Time	10 minutes
Tools	Phillips #0, flat pry tool, small pick

In plain words. The bezel is the plastic frame around the screen. It clips (and sometimes glues) onto the lid. You remove it to get to the screen, camera or display cable.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model

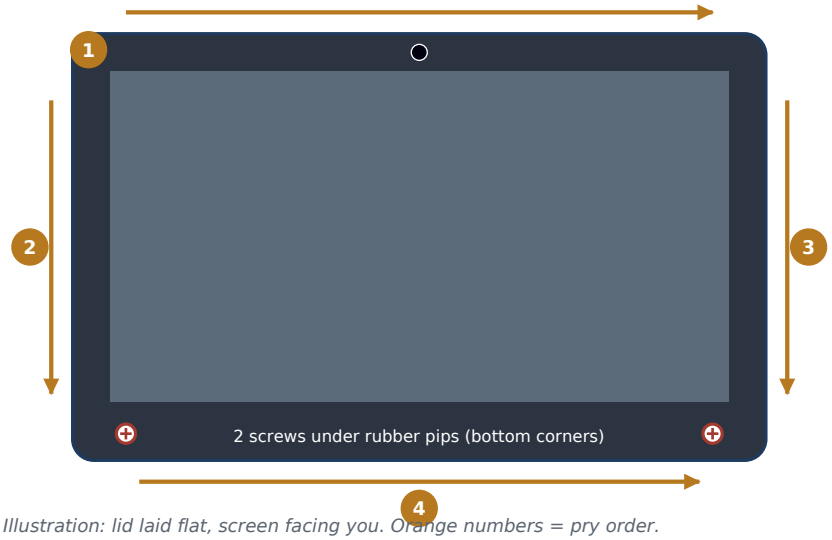


Illustration: lid laid flat, screen facing you. Orange numbers = pry order.

1. Remove the pips and the 2 bezel screws.
2. Start at the top-left corner with a flat pry tool.
3. Work down the left side, across the top, down the right side.
4. Finish along the bottom (hinge) edge last.
5. Don't pull on the camera window area.

Step by step

1. ☐ Power off and unplug. Open the lid flat.
2. ☐ Pick out the rubber **pips** covering the **two (2)** bezel (the plastic frame around the screen) screws, then remove both screws.
3. ☐ Pry from the **top-left corner** with a flat pry tool or spudger (a plastic pry stick).
4. ☐ Work down the left side, then the right side, toward the hinge.

5. ☐ Continue along the bottom edge to release the bezel. Lift it away.
6. ☐ To fit: press the bezel on around the edges until every clip snaps, refit 2 screws, and press the pips back in.

Common mistakes
✗ Prying against the screen (cracks it).
✗ Bending the bezel too far.
✗ Forgetting new adhesive on glued bezels.

If something goes wrong
Gap after refitting → A clip isn't seated or broke.
Camera blocked → Bezel isn't lined up.

Verify before closing the ticket

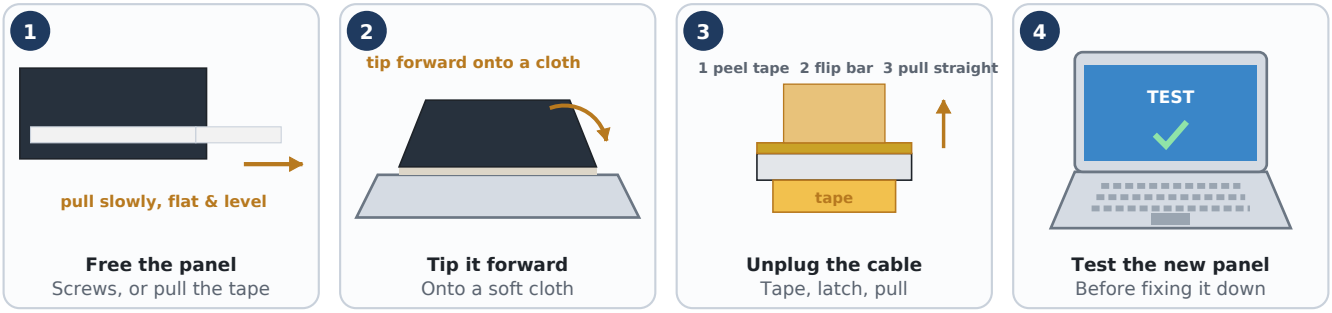
- Bezel flush all around.
- Both pips in place (students pick at loose ones).
- Lid closes fully.

R11 LCD panel replacement — LEVEL 2

Fixes	Cracked screen, dead pixels, blots, black screen not fixed by the cable
Part	NL72, NL73 and NL73-8 LCD Panel. **NL73CT needs the touch panel.**
Time	25 minutes
Tools	Phillips #0, plastic picks, safety glasses, Kapton tape

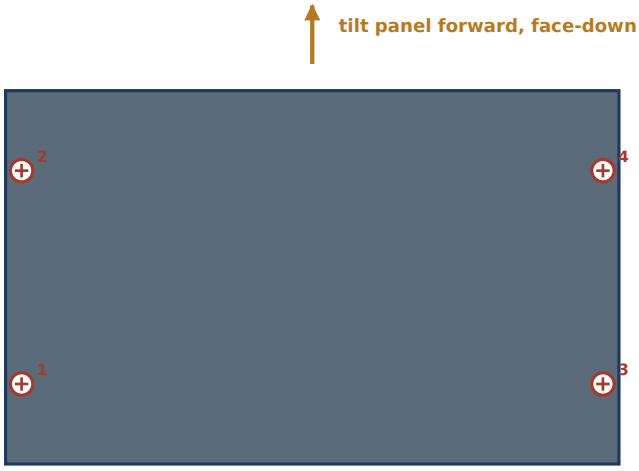
In plain words. The LCD panel is the screen itself. Once the bezel is off, it's held by small screws or stretchy tape. It tips forward so you can unplug its cable from the back.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Remove the 4 panel screws (sides of the panel).
2. Lay the panel face-down on the keyboard (cloth under it).
3. On the back: peel the tape over the connector, open the clip, pull the cable straight out.
4. Connect the new panel and TEST before screwing it down.

Illustration: lid with bezel removed. Screw positions are typical. Count them on yours.

Cracked glass. Wear safety glasses. Tape across a shattered panel before handling it.

Test first: if an external USB-C display works, the mainboard is fine and the fault is the panel or the cable.

Step by step

1. ☐ Do R3 and disconnect the battery.
2. ☐ Do R10 to remove the bezel (the plastic frame around the screen).
3. ☐ Remove the **four (4)** LCD panel screws.
4. ☐ Tilt the panel forward, face down onto a cloth on the keyboard.
5. ☐ Remove the sticky tape over the connector, open the clip, and remove the LVDS (the screen's video cable) cable from the panel.
6. ☐ Connect the cable to the new panel, close the clip, and re-tape.
7. ☐ Reconnect the battery and power on to check the picture **before** screwing the panel down.
8. ☐ Power off, disconnect the battery, and refit 4 screws, the bezel, and the D cover.

Common mistakes

✗ Pressing on the screen face.
✗ Pulling the cable before opening the latch.
✗ Ordering the wrong panel (touch vs non-touch).

If something goes wrong

New panel also black → Display cable or board: cable repair next.
Touch not working → Wrong panel type or touch cable not connected.

Verify before closing the ticket

- Full image with no lines, dead pixels, or uneven backlight.
- Brightness works across the full range.
- Image steady through full lid travel.

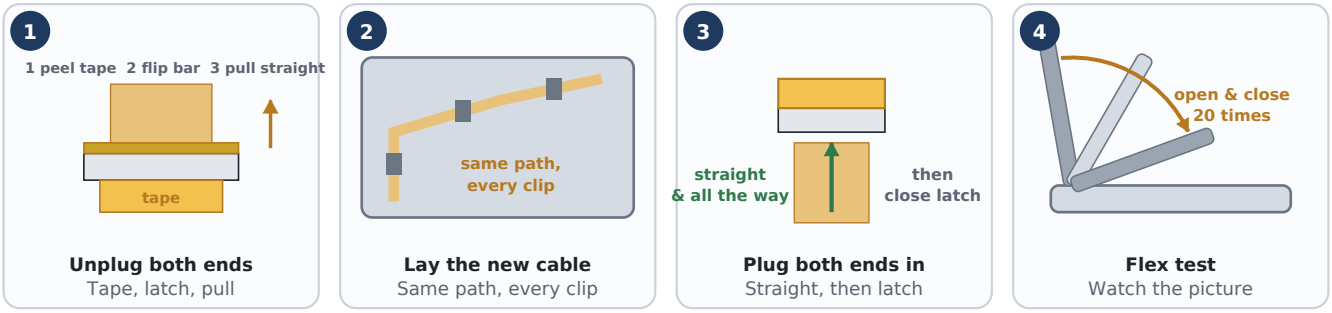
R12 LVDS (display) cable replacement

LEVEL 2

Fixes	Flicker, lines, or black screen that changes with lid angle. External display works.
Part	NL72 and NL73 LVDS Cable (Clamshell). **NL73CT uses a different cable.**
Time	40–50 minutes
Tools	Phillips #0, plastic tweezers, Kapton tape

In plain words. The display cable carries the picture from the main board up through the hinge to the screen. Hinges bend it every time the lid moves, so it wears out. Symptoms change when you move the lid.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model

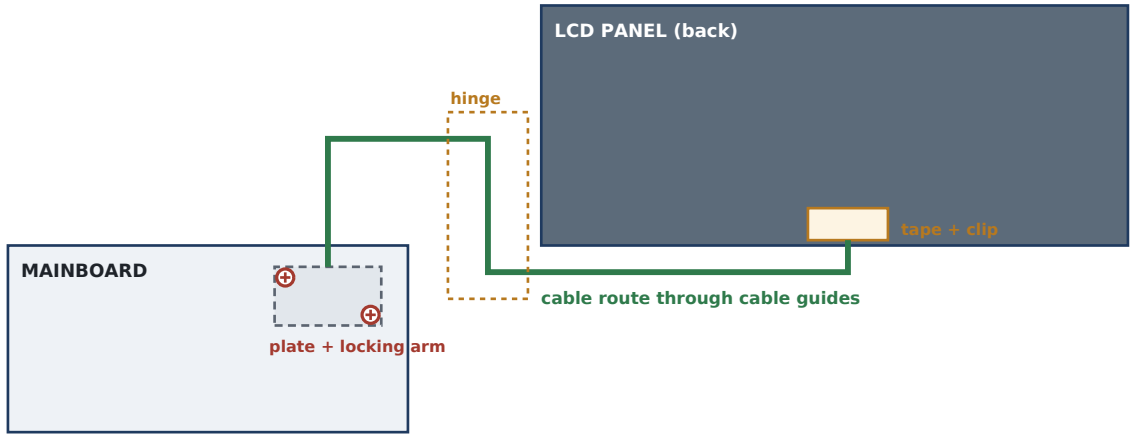


Illustration: display cable path, mainboard to panel (route simplified). Photograph the real route before removing.

Step by step

1. ☐ Do R3 and disconnect the battery.
2. ☐ At the mainboard: remove the **two (2)** screws on the protective plate, lift the locking arm, and remove the LVDS (the screen's video cable) cable.
3. ☐ Photograph the route, then unthread the cable from the cable guides.
4. ☐ Do R10 (bezel (the plastic frame around the screen)) and R11 steps 3–5 (panel face down, cable unplugged).

5. ☐ Free the cable from the lid and hinge area. Note whether the camera is wired through it.
6. ☐ Route the new cable exactly along the old path, threaded through every guide.
7. ☐ Connect both ends: panel clip closed and taped, mainboard locking arm down, plate refitted with 2 screws.
8. ☐ Reconnect the battery and test before reassembling.

Common mistakes
✗ Routing the cable a new way (it gets pinched).
✗ Leaving the tape off the connector.
✗ Not testing at different lid angles.

If something goes wrong
Flicker only at some angles → Cable still pinched or not seated.
Black screen, external display works → Panel: LCD panel repair.

Verify before closing the ticket

- Steady image at every angle. Open and close the lid 20 times while watching.
- Camera still works.

First-teardown notes — fill in the first time you do this repair

Does the camera connect through this cable? (Y/N) _____

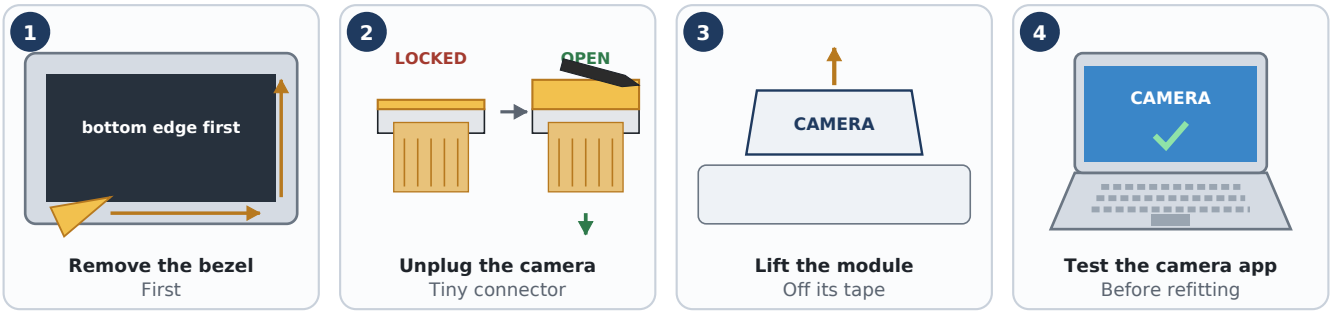
Which hinge does the cable pass through? _____

R13 Camera module replacement LEVEL 2

Fixes	Camera not detected, black image, broken lens
Part	Camera Module NB00430
Time	20 minutes
Tools	Plastic picks, tweezers

In plain words. The camera is a small strip at the top of the screen. It's usually held by sticky tape and connected by a tiny cable. Behind the bezel, it lifts out easily.

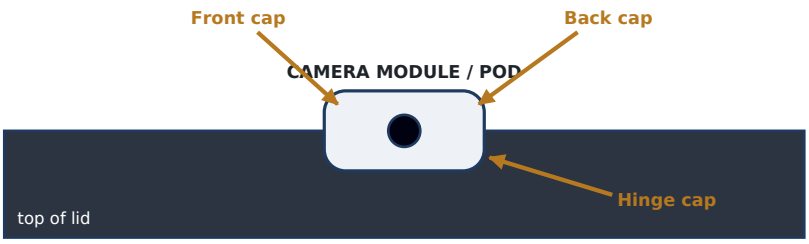
At a glance



Remove the bezel First
Unplug the camera Tiny connector
Lift the module Off its tape
Test the camera app Before refitting

The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Bezel off first (it covers the camera's front).
2. Unplug the small camera cable with tweezers.
3. Lift the module by its board edges, never the lens.
4. Test in the Camera app before closing up.

Illustration: camera at the top-center of the lid. Cap shapes simplified.

Check first: S1, a powerwash (S3), and the Camera app in both pod positions. CTL's manual lists the camera as self-repairable.

Step by step

1. ☐ Do R3 and disconnect the battery.
2. ☐ Do R10 to remove the bezel (the plastic frame around the screen).
3. ☐ Remove the camera caps needed to free the pod (R2). Record which ones.
4. ☐ Disconnect the camera cable with tweezers and lift the module out by its board edges.
5. ☐ Fit the new module in the same orientation and reconnect the cable.
6. ☐ Reconnect the battery and test the Camera app **in both pod positions** before refitting caps and bezel.

Common mistakes
✗ Touching the lens.
✗ Forgetting the privacy shutter is closed.
✗ Bending the camera strip.

If something goes wrong
Still not detected → Check the cable; try S3.
Image blurry → Lens dirty or fingerprinted.

Verify before closing the ticket

- Clear image both ways.
- Pod rotates smoothly and holds position.
- Works in a Google Meet test call.

First-teardown notes — fill in the first time you do this repair	
Caps removed to free the camera	_____
Camera cable route	_____

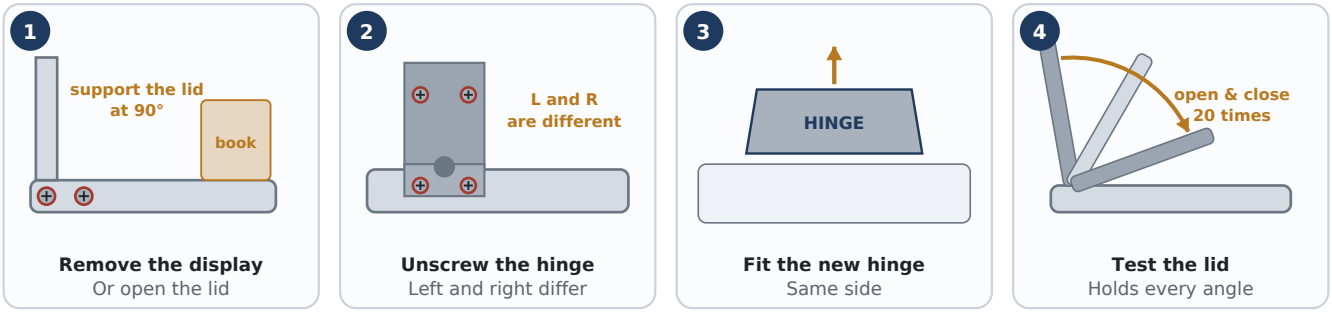
R14 Hinge replacement / removing the top half

LEVEL 2

Fixes	Floppy lid, cracked hinge, grinding. Removing the top half is also used for R8, R15, R17.
Part	Left Hinge or Right Hinge NL72/CT/L/NL73/CT
Time	30–45 minutes
Tools	Phillips #0/#1, plastic pry tool

In plain words. Hinges are metal brackets that hold the lid to the base and keep it at any angle. A cracked or loose hinge makes the lid floppy.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model

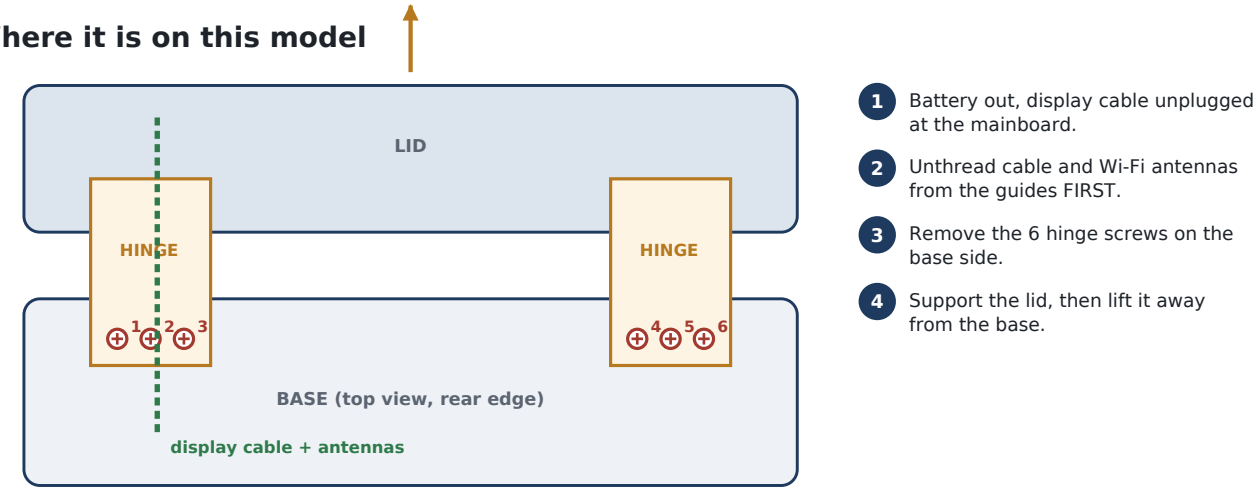


Illustration: hinge area seen from above with the D cover off. Positions approximate.

Check first: tighten the hinge screws. If a screw spins without tightening, the post is stripped. That's a cover replacement, not a hinge.

Step by step

1. ☐ Do R3 and remove the battery (CTL's prerequisite for removing the top half).
2. ☐ Disconnect the eDP (the screen's video cable)/LVDS (the screen's video cable) cable from the mainboard connector (R12 step 2).

3. ☐ Unthread the display cable and the Wi-Fi antennas from their cable guides. **CTL warns: do this before separating the halves.**
4. ☐ Remove the **six (6)** hinge screws.
5. ☐ Holding the lid so it can't fall, open it carefully and separate it from the bottom half.
6. ☐ To replace a hinge: remove the bezel (the plastic frame around the screen) (R10) and, if needed, the panel (R11) to reach the lid-side hinge screws. Swap the hinge. Left and right are mirror images.
7. ☐ Reassemble: seat the lid on the base, refit the 6 hinge screws, reroute cable and antennas through the guides, and reconnect.

Common mistakes
✗ Mixing up left and right hinges.
✗ Pinching the display cable in the hinge.
✗ Forcing a stripped screw.

If something goes wrong
Screw spins and never tightens → Stripped post: cover replacement.
Lid floppy on one side → Wrong hinge or loose screw.

Verify before closing the ticket

- Lid holds any angle and opens to 180°.
- Display steady through full travel.
- Wi-Fi strength normal (antennas not pinched).

First-teardown notes — fill in the first time you do this repair	
Lid-side hinge screw count	_____
Which hinge do the antennas pass through?	_____

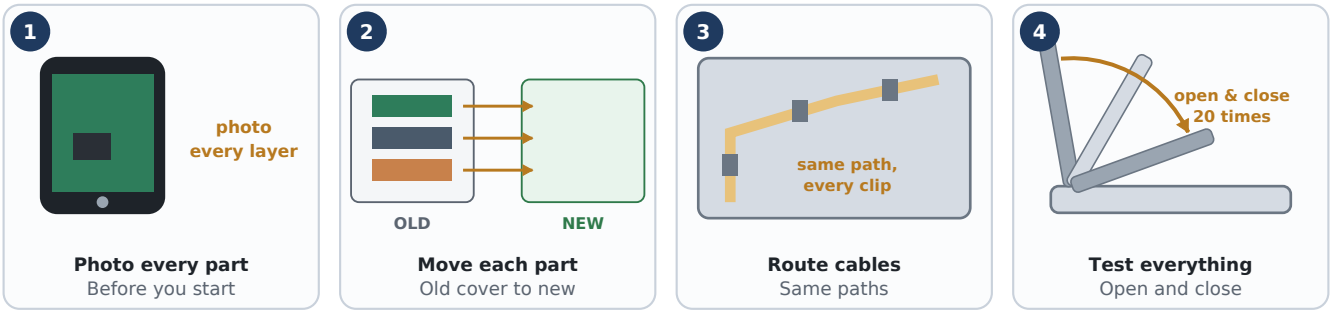
R15 A cover (lid) replacement

LEVEL 3

Fixes	Cracked or broken lid back, stripped lid hinge posts
Part	NL72/CT and NL73/CT A cover
Time	45–60 minutes
Tools	Phillips #0/#1, plastic picks, tweezers, Kapton tape

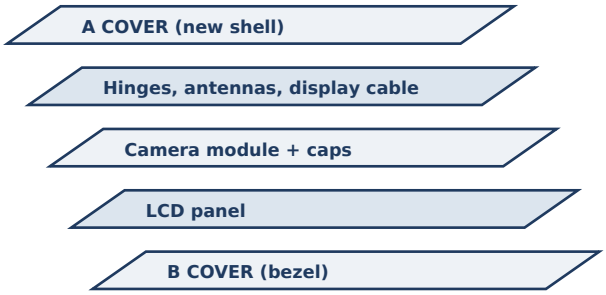
In plain words. The A cover is the back of the lid. Everything in the screen lid moves to the new back cover.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Strip the old lid top to bottom: bezel, panel, camera, hinges, cable, antennas.
2. Build the new lid in the reverse order.
3. Test the display and Wi-Fi before the bezel goes on.

Illustration: lid layers, outside (top) to inside (bottom).

This moves every lid part to a new shell, combining R10–R14. Do each of those on its own first.

Step by step

1. ☐ Do R14 steps 1–5 to separate the lid from the base.
2. ☐ Do R10 (bezel (the plastic frame around the screen)), R11 steps 3–5 (panel), and R13 (camera and caps).
3. ☐ Remove the lid-side hinge screws and both hinges.
4. ☐ Lift out the display cable and Wi-Fi antennas, noting their route. Transfer any magnets, bumpers, and tape. Photograph first.
5. ☐ Build the new A cover in reverse: antennas, cable, hinges, camera, panel (test it), bezel.
6. ☐ Reattach the lid to the base (R14 step 7).

Common mistakes
✗ Losing track of small screws.
✗ Pinching cables.
✗ Forgetting the antennas.

If something goes wrong
Wi-Fi weak → Antenna not reattached.

Verify before closing the ticket

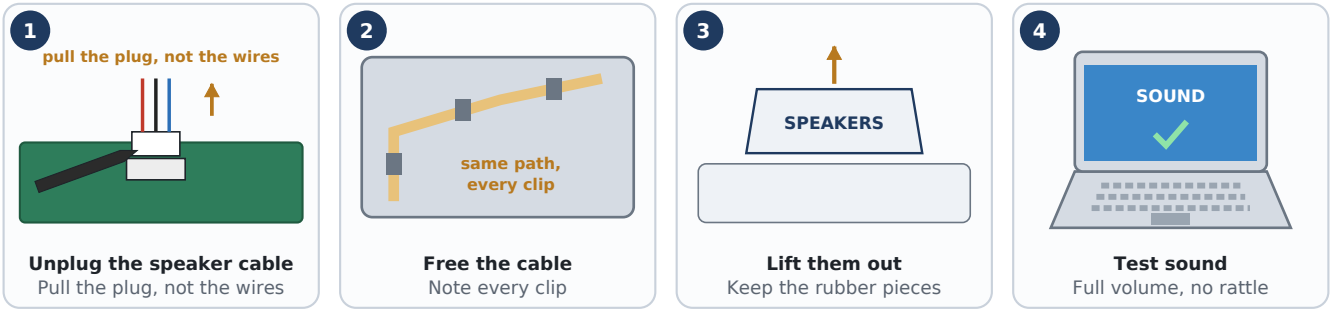
- Display, camera, and Wi-Fi all work.
- Lid closes flush. Closing the lid puts the device to sleep (magnet in place).

R16 Speaker replacement — LEVEL 2

Fixes	No sound or crackling from one or both speakers after S1 and S3
Part	Renewed Left / Right Speaker for NL72 (confirm NL73 fit with CTL)
Time	20 minutes
Tools	Phillips #0, plastic tweezers

In plain words. The speakers are two small boxes joined by one cable. They unplug from the board, unroute from a few clips, and lift out. The rubber pieces stop them rattling, so keep them.

At a glance



Unplug the speaker cable
Pull the plug, not the wires

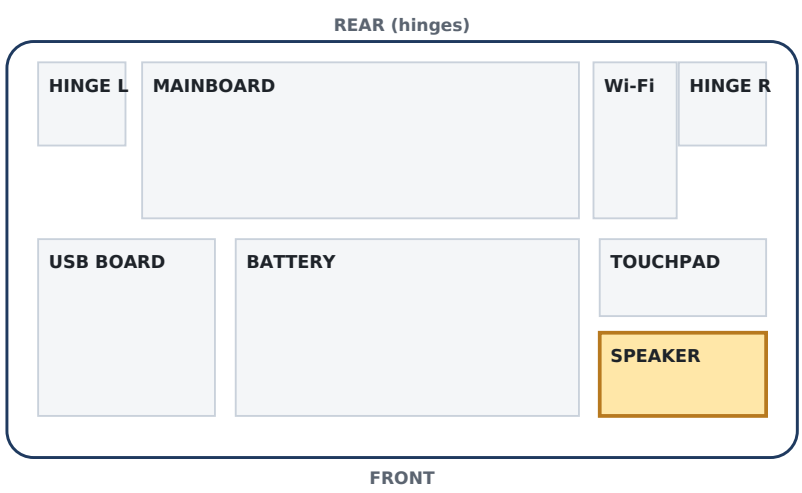
Free the cable
Note every clip

Lift them out
Keep the rubber pieces

Test sound
Full volume, no rattle

The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Battery disconnected.
2. Unplug the speaker cable at the mainboard.
3. Unroute the cable, noting each clip.
4. Lift the speaker; keep its rubber gasket.

Illustration: base interior, speaker highlighted. Position illustrative.

Check first: headphones work. If headphones also have no sound, the fault is software or mainboard, not the speakers.

Step by step

1. ☐ Do R3 and disconnect the battery.
2. ☐ Find the speakers and trace the speaker cable to the mainboard.
3. ☐ Disconnect the speaker cable at the mainboard.
4. ☐ Unroute the cable and lift the speaker(s) out. Note any rubber gaskets.
5. ☐ Fit the new speaker(s) with gaskets in place, route the cable the same way, and reconnect.
6. ☐ Reconnect the battery and test.

Common mistakes

✗ Losing the rubber isolators (speakers buzz).
✗ Pinching the cable under the cover.
✗ Replacing speakers when headphones are silent too.

If something goes wrong

Headphones also silent → Software or board: S1, S3, then lead tech.
Rattle at high volume → Missing isolator or loose screw.

Verify before closing the ticket

- Clear sound from both sides with no rattle.
- Headphone jack still works.

First-teardown notes — fill in the first time you do this repair

Speaker screws or clips? _____

Are left and right separate parts? _____

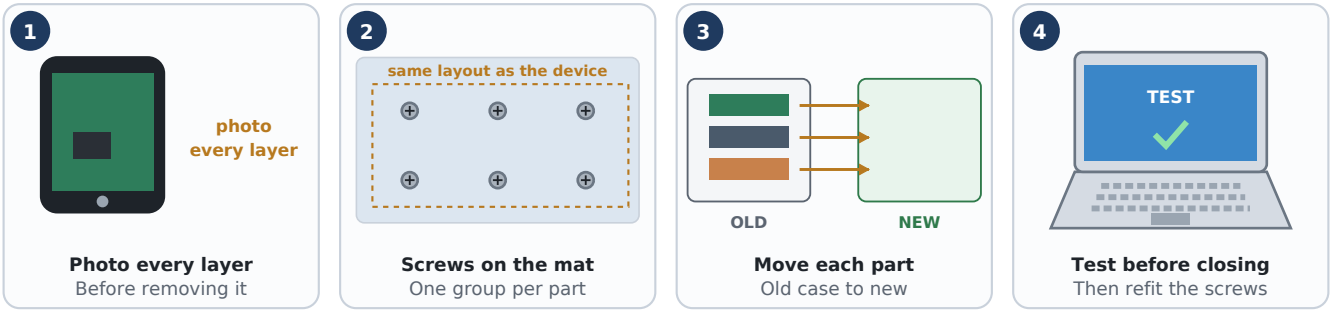
R17 Top assembly (keyboard) replacement

LEVEL 3

Fixes	Multiple dead keys, a dead row, liquid-damaged keyboard, cracked palm rest
Part	NL73 Clamshell Series keyboard, NBKEYNL73 (NL73, NL73-8, NL73CT)
Time	60–90 minutes
Tools	Phillips #0/#1, plastic tools, tweezers, ESD strap, screw mat, Kapton tape

In plain words. On this model the keyboard is built into the top case (the palm rest). To replace it, every other part moves from the old case to the new one. It's a big job: take photos and go slowly.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



Illustration: what moves to the new top assembly. Check what comes pre-fitted first.

CTL's manual: the keyboard is part of the top assembly and can't be taken apart. If it fails, replace the whole top assembly. That means moving every part into the new one.

Check the box first. CTL's listing says "keyboard." Confirm when it arrives that it's the full top assembly (C cover with keyboard), and note what's pre-fitted (touchpad? speakers?).

Step by step

1. ☐ Do R3 and remove the battery (R4).
2. ☐ Remove the Wi-Fi card (R6). CTL lists this as a prerequisite for the mainboard.
3. ☐ Separate the lid from the base (R14 steps 2–5).
4. ☐ Remove the mainboard (R19 steps 3–7).

5. ☐ Remove the USB board (R8), speakers (R16), and touchpad (R7) if not pre-fitted on the new assembly.
6. ☐ Transfer everything to the new top assembly in reverse order: touchpad, speakers, USB board, mainboard, Wi-Fi, battery.
7. ☐ Reattach the lid (R14 step 7) and reconnect every cable, checking against your photos.
8. ☐ Reconnect the battery and test everything before refitting the D cover.

Common mistakes
✗ Losing track of which screw goes where.
✗ Forgetting a small part (tape, bracket, rubber feet).
✗ Closing up before testing everything.

If something goes wrong
Something doesn't work after → Check the photo of that area: a cable isn't plugged in.
Leftover screws → Find where they go before closing.

Verify before closing the ticket

- Every key passes the S2 keyboard test.
- Touchpad, display, camera, all ports, speakers, and Wi-Fi work.
- No leftover screws.

First-teardown notes — fill in the first time you do this repair

Parts pre-fitted on new assembly	_____
Total time for the first one	_____

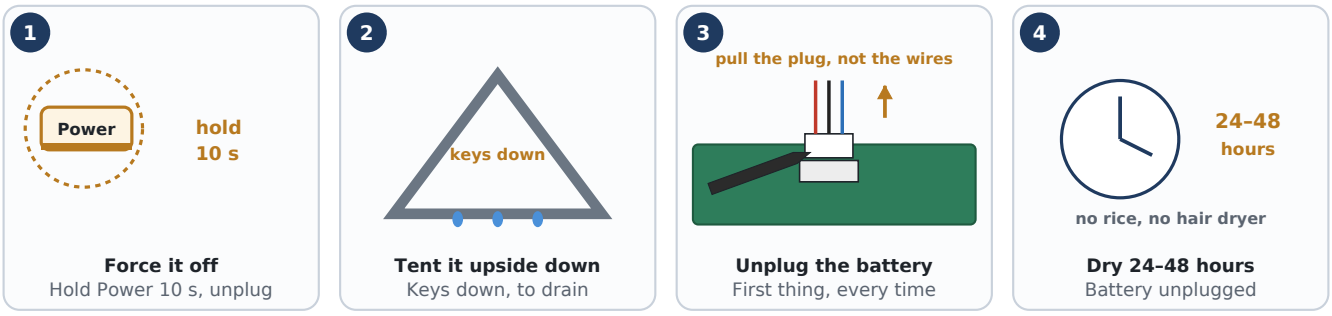
R18 Liquid spill response

LEVEL 3

Fixes	Any liquid on or in the device
Part	Often none. Sometimes top assembly or mainboard.
Time	First response 10 min, drying 24–48 hours, cleaning 30–60 min
Tools	90%+ isopropyl alcohol, soft brush, lint-free cloths

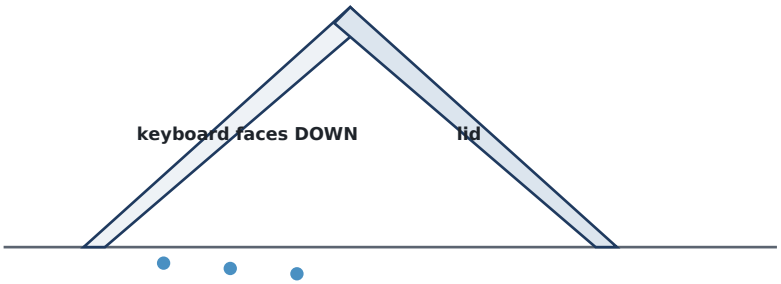
In plain words. Water and electricity don't mix. The goal is to cut the power fast, drain the liquid, and let it dry completely before it's ever turned on again. Speed matters more than anything.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Power off (hold 10 s), unplug.
2. Tent upside down so liquid drains OUT.
3. Open and disconnect the battery ASAP.
4. Dry 24–48 h. No hair dryer, no rice.

Illustration: drain position after a spill.

The NL73 keyboard is spill-resistant, but that won't protect against a full cup. Speed matters.

Do not power on to test. Current through liquid causes the corrosion that kills boards.

Step by step

1. ☐ Hold Power for 10 seconds to force off. Unplug.
2. ☐ Tent the device upside down (lid at 90°, keyboard down) to drain it.
3. ☐ Do R3 and disconnect the battery as soon as possible.
4. ☐ Blot visible liquid. Don't wipe it around.
5. ☐ Sugary or sticky spills: clean affected areas with a brush and isopropyl alcohol.
6. ☐ Dry open, battery disconnected, for 24 hours minimum (48 if soaked). No hair dryer, no rice.
7. ☐ Inspect for white or green corrosion. Clean connectors (R5).
8. ☐ Reconnect the battery, plug in, and run S2 fully.

Common mistakes

✗ Turning it on "just to check".
✗ Using rice or a hair dryer.
✗ Wiping liquid around instead of blotting.

If something goes wrong

White or green crust on the board → Corrosion: lead tech, Section 6.
Keys sticky afterwards → Keyboard repair.

Verify before closing the ticket

- All S2 tests pass.
- Sticky or dead keys: R17.
- Corrosion on the mainboard: R19 or Sec. 6.

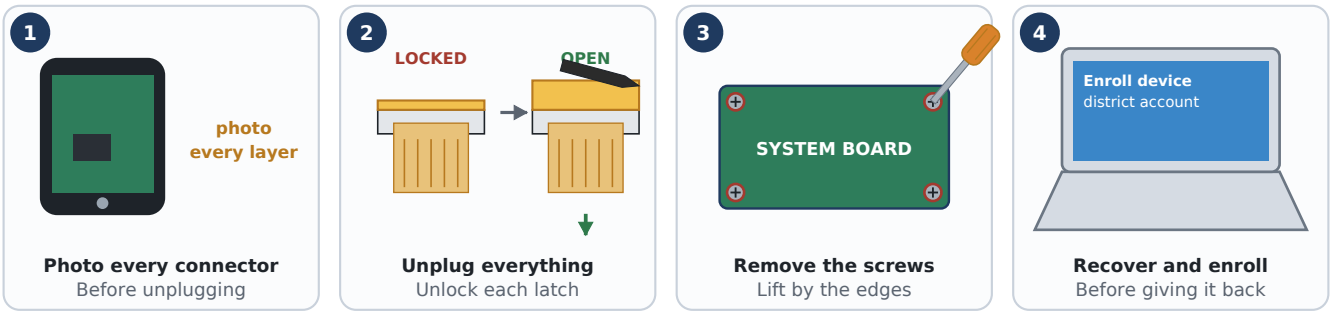
R19 Mainboard replacement

LEVEL 3

Fixes	Dead USB-C ports, no power with a good battery, failed eMMC, RAM errors, failed recovery
Part	Mainboard NL73 4/64, NB00489 • NL73-8 / NL73CT 8/64. **Match the RAM: NL73 = 4 GB, NL73-8 = 8 GB.**
Time	45–60 minutes
Tools	Phillips #0/#1, plastic tools, ESD strap

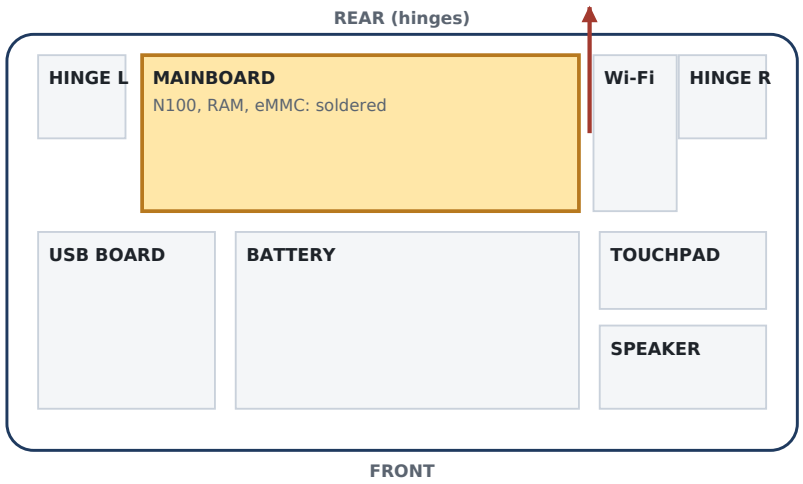
In plain words. The system board (main board) is the brain: processor, memory and storage are soldered onto it. Replacing it means unplugging everything, swapping the board, and setting the Chromebook up again. Only after every simpler fix has failed.

At a glance



The pictures show the idea, not your exact model. The numbered steps below have the details.

Where it is on this model



1. Battery and Wi-Fi card out first.
2. Unplug: 30/40-pin USB FFCs, touchpad FFC, keyboard FPC, speaker, LVDS (2-screw plate).
3. Lift from the RIGHT (Wi-Fi) side, per CTL.
4. Lift by the edges onto the ESD mat.

Illustration: base interior, mainboard highlighted. Red arrow = lift side.

Identity and enrollment. Storage and the device's identity live on the mainboard. Before swapping, the lead tech should update the asset record, deprovision the old device in the Google Admin console if needed, and plan to re-enroll. Ask CTL whether the replacement ships with the serial number set.

Confirm first with S1, S2, S3, and R5.

Step by step

1. ☐ Do R3 and remove the battery (R4).
2. ☐ Remove the Wi-Fi card (R6).
3. ☐ Disconnect the 30-pin and 40-pin USB board FFCs from the mainboard (unlock first).
4. ☐ Disconnect the touchpad FFC (flat ribbon cable), keyboard FPC, and speaker cable.
5. ☐ Remove the **two (2)** screws on the LVDS (the screen's video cable) protective plate, lift the locking arm, and remove the LVDS cable.
6. ☐ Check every cable is disconnected. **CTL's caution: disconnect all FFCs, FPCs, and cables first.**
7. ☐ Lift the mainboard from the **right (Wi-Fi) side** and remove it from the top assembly.
8. ☐ Fit the new board, then reverse steps 2–5 and refit the battery.
9. ☐ Power on. Run recovery (S3) if the board arrives blank, then enroll. Old board goes to e-waste.

Common mistakes
✗ Replacing the board when a cable was the problem.
✗ Forgetting to move parts that stay (heat sink, SSD).
✗ Not updating the asset record.

If something goes wrong
Still dead with new board → Check the battery and charger first.
Won't enroll → Lead tech checks the Admin console.

Verify before closing the ticket

- Device boots, enrolls, and passes all of S2.
- Asset record and Admin console show the correct serial.

6 Out of scope: escalate

These need soldering or board-level work, or are explicitly not self-repairable per CTL's manual ("components not described in the disassembly section are not self-repairable"). Tag the device with the ticket and send it to the lead tech.

Fault	Why	Route
USB-C port on the mainboard loose or dead	Soldered to the mainboard	R19 swap or CTL RMA
Charging circuit failure (good battery, charger, USB board)	Mainboard components	R19 or CTL RMA
eMMC storage or RAM failure	Soldered	R19 or CTL RMA
Corrosion on the mainboard	Board-level cleaning or rework	R19 or CTL RMA
Burn marks, melted plastic, burnt smell	Fire hazard	Lead tech immediately
Stripped screw posts in top assembly or lid	Cover replacement or inserts	R15 / R17 or CTL
"Security module not working" during recovery	TPM/firmware fault	CTL support
Any repair needing a soldering iron or hot air	Program policy	CTL RMA

CTL contacts

- **CTL support:** 800-642-3087 • support@ctl.net • support.ctl.net
- **Parts:** ctl.net/collections/chromebook-parts
- **Manuals and videos:** support.ctl.net, "Chromebook Repair Videos and Manuals." Has an NL73 video folder and the NL72 manual (the NL73 reference). The NL73TW videos are for the convertible but show the same general technique.

7 Sources

Checked October 2026. Screw counts and step order are from CTL's NL72 manual. The procedures are the guide author's adaptation, not CTL-issued NL73 instructions.

- CTL, "NL72 Self-Repair Manual" (Google Doc linked from CTL support; CTL's designated reference for the NL73) — tools, pre-disassembly, screw counts, procedures, recovery messages.
- CTL Support, "Chromebook Repair Videos and Manuals" (updated Nov 20, 2025). support.ctl.net/hc/en-us/articles/11785795948307
- CTL, NL73 Series spec sheet (go.ctl.net/hubfs/Flyers/04.16_CTL_NL73_7-page_v03.pdf) — CPU, RAM, display, ports, wireless, weight, SKUs, update date.
- CTL, "Introducing the CTL Chromebook NL73 Series." ctl.net/pages/ctl-chromebook-nl73-series
- CTL, "CTL Genuine Chromebook Parts" catalog pages 1–11 — part names and compatibility. ctl.net/collections/chromebook-parts
- CTL product pages: NL73 battery (NBBNL73), NL73 Clamshell keyboard (NBKEYNL73), NL73 4/64 mainboard (NB00489), camera module (NB00430).